

LECTURE NOTE ON MEC 314. INSTRUMENTATION AND CONTROL

INTRODUCTION

Measurement can be said to be the process or act of finding the size, quantity or degree of something it gives us a repeatable and dependable way of quantifying the world in which we live. It is essentially the act of comparing some unknown value with a value which is assumed to be known (otherwise known as a standard). Accurate measurement of quantities is very important in engineering. Instrumentation is when a set of instruments are used for a particular purpose (measurement or control). There are in engineering three ways in which instrumentation is used for making measurement.

- 1) Obtaining data for some events or item: this could for instance be the marking out of an item for machining and involve measurement of length and angles.
- 2) Inspecting an item to see if it matches the specification: in production engineering measurement are often concerned with determining whether an item being produced has the right dimension, shape, electrical resistance etc. that have been specified for the item.
- 3) Making measurement to ensure that a process is kept under control: many industrial processes are continuous. The purpose of measurement in this case is to ensure the proper control of the process.

Measurement systems (instruments), all measurement systems consist essentially of three main parts:

- 1) The sensing element: This frequently called the transducer is the first element. It is in some way in contact with what is being measured and produces a signal which is related to the quantity being measured. Sensing elements take information about the thing being measured and change it into some form which enables the rest of the measurement system to give a value to it. For example, if a spring balance is used to measure weight, the sensing action can be considered to be the change in length as a result of the weight. Thus, the spring balance takes information about the change changes, into a change in length of a spring.

- 2) The signal converter: The output of the sensing element then passes through a second element before reaching the display. This second element can take many forms. In general, it can be considered that the signal from the sensing element is converted into a form which is suitable for the display or control element. An example of this might be an amplifier which takes a small signal from the sensing element and makes it big enough to activate a display.

- 3) The display: The third part of the measuring system could be the display or a control system. The display element is where the output from the measuring system is displayed. This may, for example, be a pointer moving across a scale. The display element takes the signal from the signal converter and presents it in a form which enables an observer to recognize it. The control system is where the output is used to control a process.

- iii) Manual and automatic instruments
- iv) Analogue and digital instruments
- v) Self operated and power operated instruments
- vi) Self-contained and remote indicating instruments

1) Mechanical, Electric and Electronic Instruments

a) Mechanical instruments

The first instruments were mechanical in nature and the principles on which these instruments worked are even in vogue today. The earliest scientific instruments used the same three essential elements as our modern instruments do. These elements are a detector, an intermediate transfer device and an indicator, recorder or a storage device.

These instruments are very reliable for static and stable conditions. There is a large number of possibilities of mechanical instruments. It could be calipers, micrometers, scales, measuring tapes, etc. for measuring distances; a pressure gauge for measuring pressure, strain gauges for measure how much a part is stretched or compressed when a load is applied, tachometer for measuring the rotational speed, millimeter for measuring electrical voltages and currents.

However the mechanical instruments suffer from a disadvantage that they are unable to respond rapidly to measurements of dynamic and transient conditions. These instruments have several moving parts that are rigid, heavy and bulky and consequently have a large mass. The mass involved in dynamic measurements. Another disadvantage of mechanical instruments is that most of them are a potential source of noise and cause pollution of silence.

Mechanical instruments are simple in design and application. They are more durable and relatively cheaper. No external power source is required for the operation of mechanical instruments. They are quite reliable and accurate for measurements under stable conditions.

b) Electrical instruments

Electrical methods of indicating and transmitting the output are faster than the respective mechanical methods. However, an electrical system normally depends upon a mechanical pointer movement as an indicating device. Thus, owing to the inertial of mechanical movements these instruments have a limited time and frequency response. For example, some electrical recorders can give full scale response in 0.2 seconds; while the majority of industrial recorders have response time of 0.5 to 24 seconds. Some of the galvanometers can follow 50 Hz variations, but as per present day requirements of fast measurements these are also considered to be slow.

Electrical instruments are light and compact. Amplification produced is greater than that produced by mechanical means. They provide greater flexibility and are lighter in construction. These instruments consume less power and hence cause lesser load on the system.

c) Electronic instruments

Majority of the modern scientific and industrial measurements require very rapid responses. The mechanical and electrical instruments and systems cannot fulfil these requirements. There is a requirement of decreasing the response time and also the detection of dynamic changes in certain parameters. The monitoring time could be of the order of milli seconds (ms) and many a times, micro seconds. This has led to the design of today's electronic instruments and their associated circuitry. These instruments involved vacuum tubes or semi-conductor devices. The present-day practice is to use semi-conductor devices owing to their many advantages over their vacuum tube counterparts. Since in electronic devices the only movement involved is that of electrons and the inertia of electrons being very small, the response time of these devices is extremely small. For

example, a C.R.O. is capable of following dynamic and transient changes of the order of a few nano seconds (10^{-9} s).

Electronically controlled power supplies are used to provide stable voltages for studies in the field of chemical reactions and nuclear instrumentation. Electronic instruments are steadily becoming more reliable on account of improvements in design and manufacturing processes of semiconductor devices. Another advantage of using electronic devices is that very weak signals can be detected by using pre-amplifiers and amplifiers. The foremost importance of the electronic instruments is the power amplification provided by the electronic amplifiers. Additional power may be fed into the system to provide an increased power output beyond that of the input. This has been only possible through the use of electronic amplifiers, which have no important mechanical counterpart. This is particularly important where the data presentation devices use stylus type recorders, galvanometers, cathode ray oscilloscopes and magnetic tape recorders.

It is a fact that hydraulic and pneumatic systems may be used for power amplification of signals. However, their use is limited to slow acting control applications like servo-systems, chemical processes and power systems. Electronic instruments find extensive use in detection of electromagnetically produced signals such as radio, video, and microwave. Electrical and electronic instruments are particularly useful in the intermediate signal modifying stage. Electronic instruments are light compact and have a high degree of reliability. Their power consumption is very low.

Electronic instruments make it possible to build analogue and digital computers without which the modern developments in science and technological are virtually impossible. Computers require a very fast time response and it is only possible with use of electronic instruments. The mathematical processing of signal, such as, summation, differentiating and integrating is possible with electronic measurements. With these instruments non-contact or remote measurements are also possible.

ii) Absolute/primary and Secondary Instruments

Electrical measurements of different parameters like current, voltage, power, energy, etc. are most essential in any industry. These are among the oldest of all measurements. The various electrical instruments may be broadly divided into two categories:

- 1) Absolute instruments
- 2) Secondary instruments

a) Absolute/primary instruments

Absolute/primary instruments are those which give the value of electrical quantity to be measured in terms of the constants of the instruments and their deflection only e.g. tangent galvanometer. These instruments are rarely used except in standard laboratories, especially for calibration of secondary instruments.

b) Secondary instruments

Secondary instruments are those in which the values of electrical quantity to be measured can be determined from the deflection of the instruments only when they have been pre-calibrated by comparison with an absolute instrument. Without calibration, the deflection of such instruments is meaningless.

Working with absolute instruments for routine work is time consuming since every time a measurement is made, it takes a lot of time to compute the magnitude of the quantity under use of the absolute instruments being merely confined within laboratories as standardizing instruments. A voltmeter, a glass thermometer and a pressure gauge are typical examples of secondary instruments.

Secondary type of measuring instruments has been classified in the following categories:

Indicating instruments

Indicating instruments are those which indicate the instantaneous value of the variables being measured, at the time at which it is being measured. Their indications are given by pointers moving over calibrated dials or scales, e.g., ammeter, voltmeter and wattmeter. This movement of pointer and deflection is not constant but depends on the quantity it measures. As the needle deflects instruments indicate the amount of current, voltage or any quantity, these are called deflection type of instruments.

Recording instruments

Recording instruments are those which give a continuous record of variations of the measured variable over a selected period of time. The moving system of the instrument carries an inked pen which rests tightly on a graph chart. These instruments will go on recording on a graph sheet fixed on the instrument all the variations of the quantity in the time it is connected in the circuit. Normally these recordings will be for one day and the recorded sheets are kept as a record of variation of the quantity with time.

Integrating instruments

These are the instruments which will add up the quantity as the time passes or in other words will give a total account of quantity spent in a given time for which it is connected in a circuit. For example, an electric meter measure and register, by a set of dials and pointers, either the total quantity of electricity (in ampere-hours) or the total amount of electrical energy (in watt-hours or kilowatt-hours) supplied to a circuit over a period of time and are known as ampere-hour meters, watt hour meters, energy meters, etc. Another example is house hold water meter. Deflecting type instruments are again classified as follows:

- a) Depending upon working principle, such as, moving coil, moving iron, dynamometer, electrostatic type, induction type
- b) Depending upon the quantity it measures, such as, voltmeter, ammeter, ohm meter, power factor meter, energy meter etc.
- c) Depending upon the shape of the instruments, such as, portable, panel board type with flush mounting or surface mounting.

Deflection is normally with in 90° , but circular scale instruments are also available which give about 250° deflection. All the deflecting instruments are marked on scale to indicate its working principle by symbols.

iii) Manual and Automatic Instruments

Manual require the services of an operator, where as in automatic instruments the operator is not required. For example, measurement of rotational speed by a hand operated tachometer an operator is required to make the contact of the instrument with the rotating shaft. For measurement of temperature by a resistance thermometer by Wheat stone bridge in its circuit an operator is required to indicate the temperature being measured. Whereas, in measurement of temperature by mercury-in-glass thermometer, no operator is required.

iv) Self Operated and Power Operated Instruments

A self-operated instrument does not require any external power source for its operation. In such instruments the output energy is supplied by the input signal e.g. a dial indicator or mercury-in-glass type thermometer.

In power operated instruments some auxiliary power source is required for its operation. This external power source could be electricity, compressed air etc. In such cases the input signal

supplies only the insignificant portion of the output power e.g. an electro-mechanical measurement system.

v) Self Contained and Remote Indicating Instruments

A self-contained instrument has all the physical elements in one assembly e.g. an analogue ammeter or a mercury-in-glass thermometer etc. Whereas, in a remote indicating instrument has primary sensory element and the secondary indicating element are located at two different locations linked by transmitting element. These locations could be long distance apart. In modern instrumentation technology such type of arrangement is quite necessary and vogue. In vii) Contact Type and Non-Contact Type Instruments

In contact type instruments the sensing element of the instrument contacts the control medium for the measurements, for example mercury-in-glass thermometer. Whereas in non-contact type instruments the sensor does not contact the control medium. The non-contact type measurement includes optical, radioactive or radiation measurements. Such as, radiation or optical pyrometer, non-touch tachometer etc.

Performance characteristics

The performance characteristics of an instrument are very necessary for choosing the most suitable instrument for a specific measuring task. This generally has two sub areas:

- static characteristics
- dynamic characteristics

Static characteristics: the static characteristics are generally considered for instrument used in measuring unvarying process condition. These characteristics are usually obtained by one form or another of a process called *calibration*.

Calibration may be defined, in general, as the process for determination, by measurement or comparison with a standard, of the correct value each scale reading on a meter or measuring instrument; or determination the settings of a control device respond to particular values of voltage, current, frequency, pressure, flow or some other output.

In performing a calibration of an instrument, the following steps are necessary

- Examine the construction of the instrument and identify and list possible inputs.
- Decide, as best as one can, which of the inputs will be significant application for which the instrument is to be calibrated.
- Procure apparatus that will allow all significant inputs to vary ranges considered necessary.
- By holding some inputs constant, varying others, and recording the developed the desired static input-output relations.

The following are some of the terms commonly used to describe the performance of measuring systems:

Accuracy: The accuracy of an instrument is the extent to which the reading it gives might be wrong or the ability of a device or a system to respond to the true value of a measured variable under reference conditions. It is usually quoted as a percentage of the full-scale deflection (f.s.d) of the instrument. Thus, for example, an ammeter might have an f.s.d. of 5 A and accuracy quoted as $\pm 5\%$. This means that 0.25 A is either added or subtracted from any reading taking on this ammeter. Accuracy can either be static for slow changing quantities or dynamic for quantities that changes quickly.

Precision: This is the degree of exactness for which an instrument is designed to perform. It consists of two characteristics, conformity and the number of significant figures to which the measurement may be made. The more the significant the figures, the greater the precision of the instrument.

Repeatability: The repeatability of an instrument is its ability to display the same reading for repeated applications of the same value of the quantity being measured.

Reliability: The reliability of an instrument is the probability that it will operate to an agreed level of performance under the conditions specified for its use.

Reproducibility: The reproducibility or stability of an instrument is its ability to display the same reading when it is used to measure a constant quantity over a period of time or when that quantity is measured on a number of occasions.

Sensitivity: This can be defined as the ratio of the change in output to the change in input which causes it, at steady state conditions. the sensitivity of an instrument is given by

$$\text{sensitivity} = \frac{\text{change in instrument scale reading}}{\text{change in the quantity being measured}}$$

Eq 1.1

Thus, for example, a volt meter might have a sensitivity of 1 scale division per 0.05V. This means that if the voltage being measured changes by 0.05V, then the reading of the instrument will change by one scale division.

Resolution: The resolution or discrimination of an instrument is the smallest change in the quantity being measured that will produce an observable change in the reading of the instrument.

Range: The range of an instrument is the limits between which readings can be made

Dead space: The dead space of an instrument is the range of values of the quantity being measured for which it gives no reading.

Threshold: If the quantity being measured is increased from zero a certain minimum level might have to be reached before the instrument responds and gives a detectable reading. This is called the threshold.

Hysteresis: Instruments can give different reading for the same value of measured quantity depending on whether that value has been reached by a continuously increasing quantity continuously decreasing change. This effect is called hysteresis and it occurs as a result of such things as bearing friction and slack motion in gears in instrument.

• Dynamic characteristics: Instruments rarely respond to changes in measured variables, rather they exhibit a characteristic of slowness or sluggishness due to things as mass, thermal capacitance, fluid capacitance or electric capacitance. In addition to this, pure delay in time is often encountered when the instrument waits for some reactions to take place. the dynamic behavior of an instrument is determined by subjecting its primary element to some unknown and predetermined variations in measured quantity. the three most common variations are:

- step change in which the primary element is subjected to an instantaneous and finite change in measured variable
- linear change in which the primary element is following a measured variable changing linearly with time
- sinusoidal change in which the primary element follows a measured variable the magnitude of which changes in accordance with the sinusoidal function of a constant amplitude.

The dynamic characteristics of an instrument as follows:

Speed of response: It is the Rapidity with which an instrumental response to changes in the measured quantity

Fidelity: It is the degree to which an instrument indicates the changes in the measured variable without dynamic error

Lag: It is retardation or delay in the response of an instrument changes in the measured variable

Dynamic error: It is the difference between the true value of a quantity changing with time and the value indicated by the instrument if not static error is assumed.

Error = measured value - true value

Eq 1.2

Sources of error: the sources of error other than the inability of the instrument to provide true measurement, are listed as follows:

- i. insufficient knowledge of process parameters and design conditions
- ii. poor design
- iii. change in process parameters, irregularities, upsets etc
- iv. poor maintenance
- v. error caused by people who operate instrument equipment
- vi. certain design limitations

Dynamic response of zero order instruments:

The relationship between any input and the output can by application of suitable simplifying assumptions, be written as:

$$a_n \frac{d^n x_o}{dt^n} + a_{n-1} \frac{d^{n-1} x_o}{dt^{n-1}} \dots + a_1 \frac{dx_o}{dt} + a_0 x_o = b_m \frac{d^m x_i}{dt^m} + b_{m-1} \frac{d^{m-1} x_i}{dt^{m-1}} + \dots + b_1 \frac{dx_i}{dt} + b_0 x_i \quad \text{Eq 1.3}$$

Where:

x_o = output quantity

x_i = input quantity

t = time

a 's, b 's = combination of system physical parameters, assumed constant.
When all the a 's and the b 's other than a_0 and b_0 of the eq 1.3 are assumed to be zero, the differential equation then degenerates into the simple algebraic equation given as:

$$a_0 x_o = b_0 x_i$$

Eq 1.4

Any instrument that closely obeys the Eq 1.4 over its intended range of operating conditions is defined as a zero-order instrument.

The static sensitivity (or steady-state gain) of a zero-order instrument may be defined as follows:

$$x_o = \frac{b_0}{a_0} x_i = K x_i$$

Eq 1.5

Where: $K = \frac{b_0}{a_0}$ = static sensitivity

Since the equation $x_o = K x_i$ is an algebraic equation, it is clear that, no matter how x_i might vary with time, the instrument output (reading) follows it perfectly with no distortion or time lag of any sort. Thus, a zero-order instrument represents ideal or perfect dynamic performance. A practical example of a zero-order instrument is the displacement measuring potentiometer.

Dynamic response of first order instruments: If in eq 1.3 above, all a 's and b 's other than a_1 , a_0 and b_0 are taken to be zero, we get:

$$a_1 \frac{dx_o}{dt} + a_0 x_o = b_0 x_i$$

Eq 1.6

Any instrument that follows the eq 1.6 is known as a first order instrument. By dividing eq by a_0 , the equation can be written as:

$$\frac{a_2}{a_0} \cdot \frac{dx_o}{dt} + x_o = \frac{b_o}{a_0} x_i$$

Or

$$(\tau D + 1)x_o = Kx_i$$

Where:

Eq 1.7

$$\tau = \frac{a_1}{a_0} = \text{time constant}$$

$$K = \frac{b_o}{a_0} = \text{static sensitivity}$$

the time constant τ always has the dimensions of time, while the static sensitivity K has the dimensions input/output. The operational transfer function of any first order instrument is:

$$\frac{x_o}{x_i}(D) = \frac{K}{D+1}$$

Eq 1.8

A very common example of a first order instrument is the mercury in glass thermometer
Dynamic response of second order system: A second order instrument is the one that follows the equation

$$a_2 \frac{d^2 x_o}{dt^2} + a_1 \frac{dx_o}{dt} + a_0 x_o = b_o x_i$$

Eq 1.9

The above equation can be reduced as

$$\left[\frac{D^2}{(\omega_n)^2} + \frac{2\xi D}{\omega_n} + 1 \right] x_o = Kx_i$$

Eq 1.10

Where:

$$\omega_n = \frac{a_0}{a_2}$$

= *undamped natural frequency in $\frac{\text{rad}}{\text{time}}$*

$$\xi = \frac{a_1}{\sqrt{a_0 a_2}} = \text{damping ratio, dimensionless}$$

$$K = \frac{b_o}{a_0} = \text{static sensitivity}$$

The operational transfer function of any second order instrument is

$$\frac{x_o}{x_i}(D) = \frac{K}{(D^2/\omega_n^2 + 2\xi D/\omega_n + 1)}$$

Eq 1.11

A good example of a second order instrument is the spring balance.

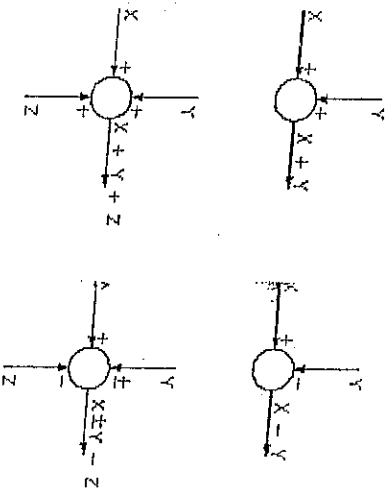
Fundamental and derived quantities

Basic elements of a block diagram Blocks Transfer functions of elements inside the blocks
Summing points Take off points **Arrow** system is a pictorial representation of the functions performed by each component and of the flow of signals. The elements of a block diagram are block, branch point and summing point.

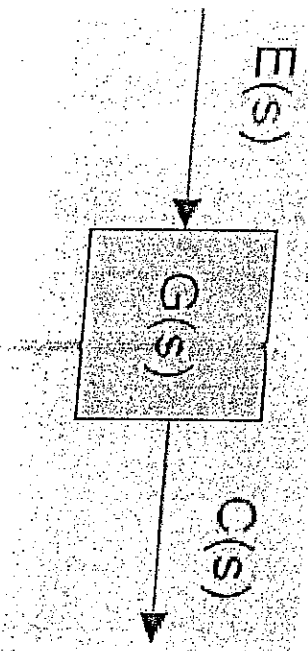
Block In a block diagram all system variables are linked to each other through functional blocks. The functional block or simply block is a symbol for the mathematical operation on the input signal to the block that produces the output.



Summing point Although blocks are used to identify many types of mathematical operations, operations of addition and subtraction are represented by a circle, called a summing point. As shown in Figure a summing point may have one or several inputs. Each input has its own appropriate plus or minus sign. A summing point has only one output and is equal to the algebraic sum of the inputs.

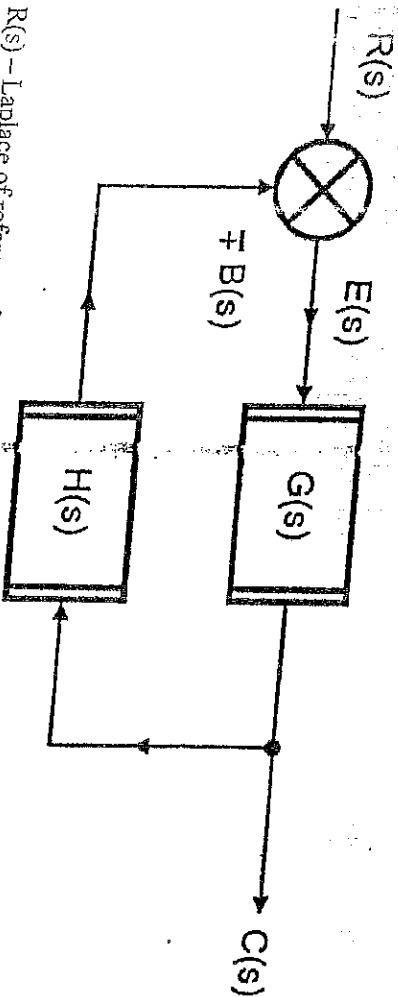


A takeoff point is used to allow a signal to be used by more than one block or summing point. The transfer function is given inside the block. The input in this case is $E(s)$. The output in this case is $C(s)$.

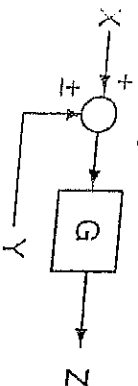


Functional block – each element of the practical system represented by block with its T.F.
Branches – lines showing the connecting between the blocks
Arrow – associated with each branch to indicate the direction of flow of signal
Closed loop system **Summing point** – comparing the different signals
Take off point – point from which signal is taken for feed back

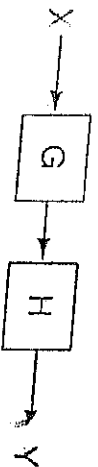
Advantages of Block Diagram Representation Very simple to construct block diagram for a complicated system. Function of individual element can be visualized. Individual & Overall performance can be studied. Over all transfer function can be calculated easily. **Disadvantages of Block Diagram Representation** No information about the physical construction. Source of energy is not shown.



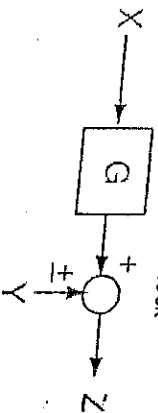
$R(s)$ – Laplace of reference input $r(t)$ $C(s)$ – Laplace of controlled output $c(t)$ $E(s)$ – Laplace of error signal $e(t)$ $B(s)$ – Laplace of feedback signal $b(t)$ $G(s)$ – Forward path transfer function
Block diagram reduction technique Because of their simplicity and versatility, block diagrams are often used by control engineers to describe all types of systems. A block diagram can be used simply to represent the composition and interconnection of a system. Also, it can be used together with transfer functions, to represent the cause-and-effect relationships throughout the system. Transfer Function is defined as the relationship between an input signal and an output signal to a device.
Block diagram rules
Cascaded blocks



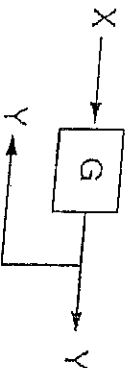
Moving a summer beyond the block



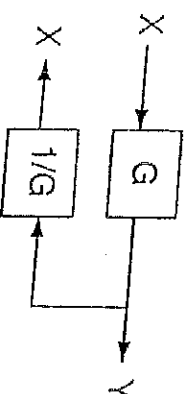
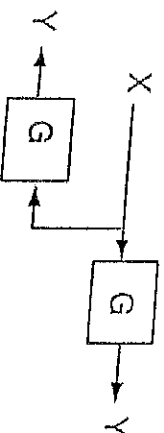
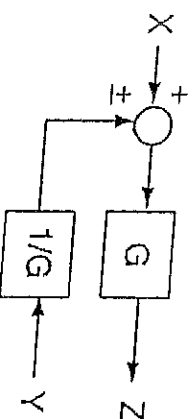
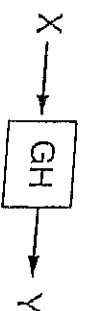
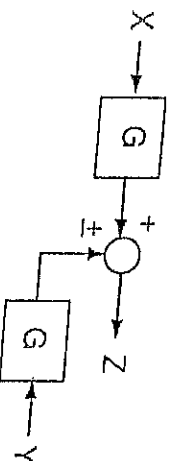
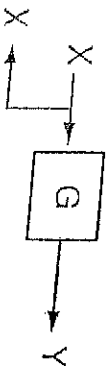
Moving a summer ahead of block



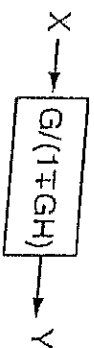
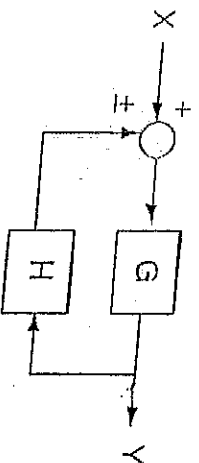
Moving a pick-off ahead of block



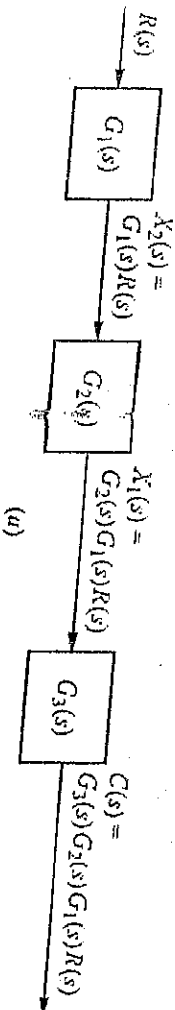
Moving a pick-off behind a block



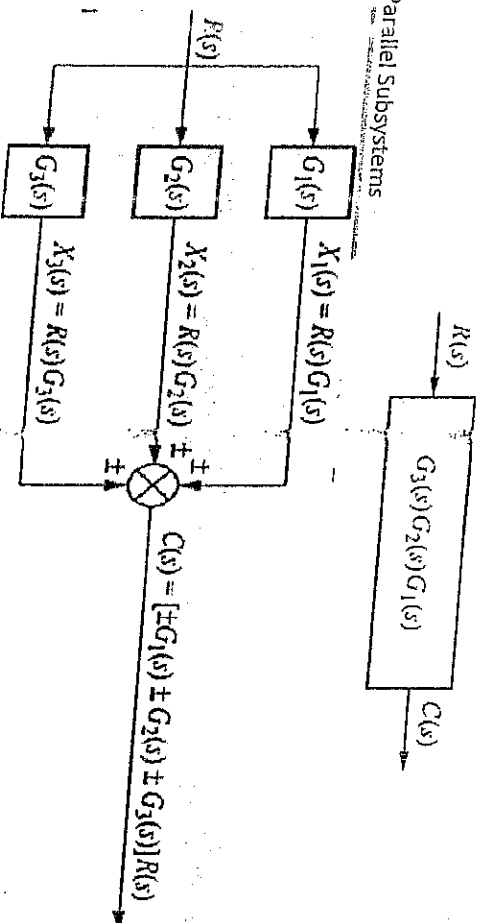
Eliminating a feedback loop



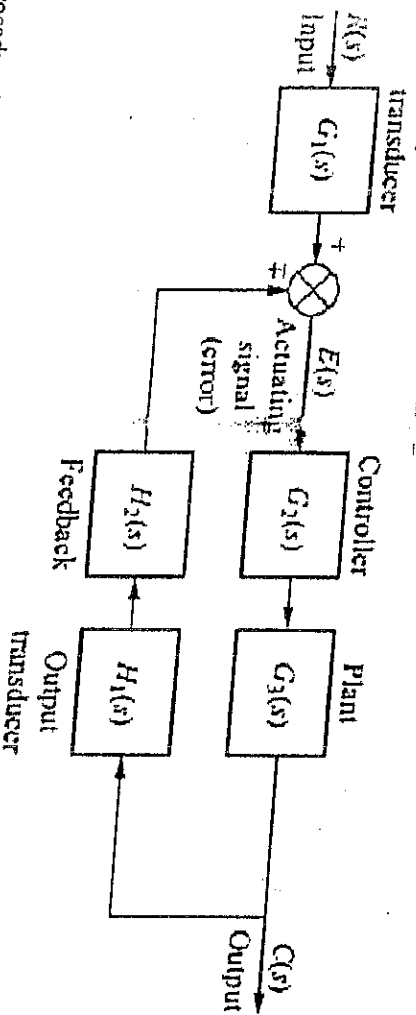
Cascaded Subsystems



Parallel Subsystems



Feedback Control System



Procedure to solve Block Diagram Reduction Problems

Step 1: Reduce the blocks connected in series

Step 2: Reduce the blocks connected in parallel

Step 3: Reduce the minor feedback loops

Step 4: Try to shift take off points towards right and Summing point towards left

Step 5: Repeat steps 1 to 4 till simple form is obtained Step 6: Obtain the Transfer Function of Overall System

Problem 1 Obtain the Transfer function of the given block diagram

Quantity		
Fundamental (Basic) Units	Unit	Unit Symbol
Length	Meter	m
Mass	Kilogram	kg
Time	Second	s
Electric Current	Ampere	A
Thermodynamic Temperature	Kelvin	K
Luminous Intensity	Candela	cd
Supplementary Units		
Plane Angle	Radian	
Solid Angle	Steradian	
Quantity of Substance	Mole	mol
Derived Units		
Area	Square metre	m ²
Volume	Cubic metre	m ³
Velocity	Metre per second	m/s
Angular Velocity	Radian per second	rad/s
Frequency	Hertz	Hz
Density	Kilogram per cubic metre	kg/m ³
Acceleration	Metre per second squared	m/s ²
Angular Acceleration	Radian per second squared	rad/s ²
Force	Newton	N (kg m/s ²)
Pressure, Stress	Newton per square metre	N/m ²
Power	Watt	W (J/s)
Work, Energy	Joule	J (Nm)
Potential difference, Electromotive force	Volt	V (W/A)
Electric Capacitance	Farad	F (A s/V)
Electric Resistance	Ohm	Ω (V/A)
Magnetic Flux	Weber	Wb (V s)
Inductance	Henry	H (V s/A)
Electric Field strength	Volt per metre	V/m
Magnetic Field strength	Ampere per metre	A/m
Luminous Flux	Lumen	lm (cd sr)
Luminance, Brightness	Candela per square metre	cd/m ²
Illumination	Lux	l x (lm/m ²)

Multiple	SI Prefix	Symbol
10^{12}	tera	T
10^9	giga	G
10^6	mega	M
10^3	kilo	k
10^2	hecto	h
10	deka	da
10^{-1}	deci	d
10^{-2}	centi	c
10^{-3}	milli	m
10^{-6}	micro	μ
10^{-9}	nano	n
10^{-12}	pico	p
10^{-15}	femto	f
10^{-18}	atto	a

Exercises:

- 1) A hydrometer is specified as having a range of 600-650 kg/m³ and an accuracy of ± 0.5 kg/m³. Explain the significance of this data.
- 2) An ammeter is quoted to have a range of 0-3 A and an accuracy of $\pm 2\%$ f.s.d. what is the accuracy which can be quoted for a current reading of 2 A?
- 3) A thermocouple is specified as having a sensitivity of 0.03 millivolt/ $^{\circ}$ C. What does this mean?
- 4) A force measuring system has a range of 0-200N with a resolution of 0.1% f.s.d. what is the smallest change in force that can be measured?

Measurement of Displacement:

Displacement can be defined as change in position between two points. Displacement measurement, both linear and angular is fundamental to all measurements. Many measurements, such as force, strain, pressure, temperature, level e.t.c. boil down to the measurement of displacement in the ultimate analysis.

In instrumentation, displacement thought of in terms of a motion of a few millimeters or less. Translational displacement transducers are instruments that measure the motion of a body in a straight line between two points. Apart from their use as a primary transducer measuring the motion of a body, translational displacement transducers are also widely used as a secondary component in measurement systems, where some other physical quantity, such as pressure, force, acceleration, or temperature, is translated into a translational motion by the primary measurement transducer. Displacement measuring devices can either be mechanical, e.g. dial indicator, float e.t.c. or electrical e.g. potentiometer, LVDT e.t.c.

The Dial Indicator

The dial indicator consists of a plunger which can move up or down moving the dial hand in either a clockwise or counter-clockwise direction. Very slight upward movement on the plunger moves it upward and the movement is indicated by the dial pointer. The dial is graduated into 100 divisions. A full revolution of the pointer about this scale corresponds to 1mm travel of the plunger. Thus, a turn of the pointer by one scale division represents a plunger travel of 0.01mm. The basic features of a dial gauge consist of a body with a circular graduated dial, a contact point connected to a gear train, and an indicating hand that directly indicates the linear displacement of the contact point. The contact point is first set against the master, and the dial scale is set to zero by rotating the bezel. Now, the master is removed and the workpiece is set below the contact point; the difference in dimensions between the master and the workpiece can be directly read on the dial scale. Dial gauges are used along with V-blocks in a metrology laboratory to check the roundness of components. A dial gauge is also part of standard measuring devices such as bore gauges, depth gauges, and vibrometers.

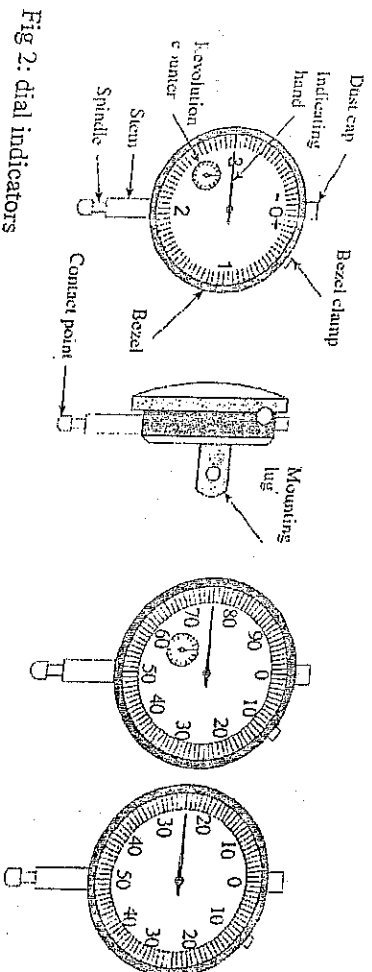


Fig 2: dial indicators

Working Mechanism of Dial Indicators

Figure 3 illustrates the mechanism used in a dial indicator in order to achieve high magnification using a set of gears and pinions. The plunger and spindle are usually one piece. The spindle measurement movement and thereby applies the necessary gauging pressure. A coil spring resists the return of the mechanism to the 'at-rest' position after each measurement. The plunger carries a rack, which meshes with a gear (marked gear A in the figure). A larger gear, B, mounted on the same spindle as gear A, rotates by the same amount and transfers motion to gear C. Attached to gear C is another gear, D, which meshes with gear E. Gear F is of the same size as gear E and is given by $TD/TE \times TB/TC$, where TD, TE, TB, and TC are the number of teeth on the gears D, E, B, and C, respectively. The magnification is further enhanced at the tip against the direction of gauging movement. This eliminates backlash that would be caused by gear wear. The gears are precision cut and usually mounted on jeweled bearings.

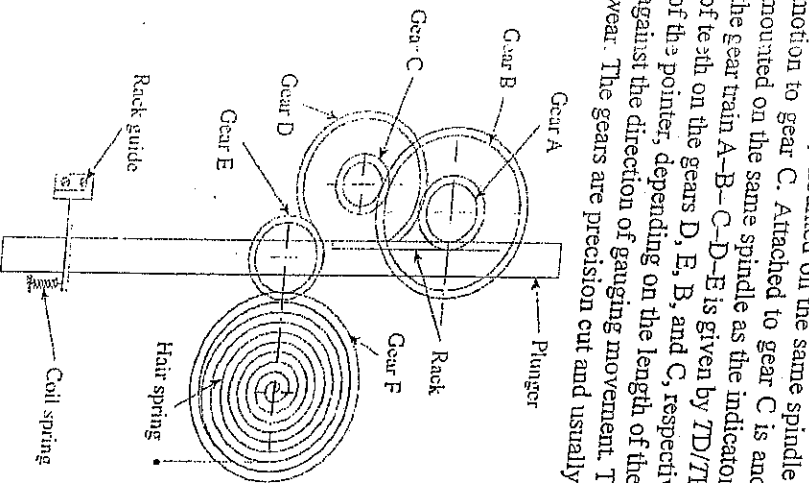


Fig 3: working mechanism of a dial indicator

Float Type Level Measurement

Float Type Level Indicator most operated level indicator is used to measure liquid levels in a tank in which a float rests on the surface of liquid and follows the changing level of liquid. The movement of the float is transmitted to a pointer through a suitable mechanism which indicates the level on a calibrated scale. Various types of floats are used such as hollow metal spheres, cylindrical-shaped floats and disc-shaped floats.

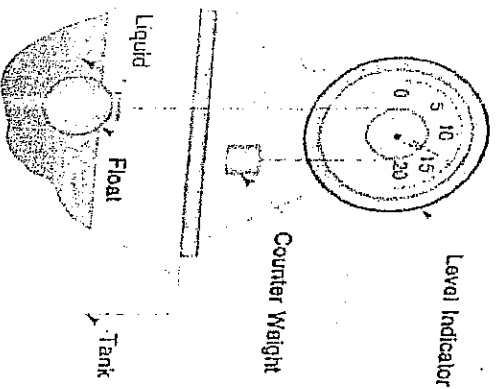


Fig 4: a float type liquid level indicator

Linear Variable Differential Transformer (LVDT):

A linear variable differential transformer (LVDT) is one of the most popular electromechanical devices used to convert small mechanical displacements (of the order of a few millimeters or fractions of a millimeter) into amplified electrical signals. LVDT, consists of a transformer with a single primary winding and two secondary windings connected in the series-opposing manner with a core of the transformer so that all motions of the body are transferred to the central iron core. For an excitation voltage V_s given by $V_s = V_p \sin(\omega t)$, the e.m.f.s induced in the secondary windings V_a and V_b are given by

$$V_a = K_a \sin(\omega t - \phi); V_b = K_b \sin(\omega t - \phi)$$

Parameters K_a and K_b depend on the amount of coupling between the respective secondary and primary windings and hence on the position of the iron core. With the core in the central position, $K_a = K_b$, we have $V_a = V_b = K \sin(\omega t - \phi)$.

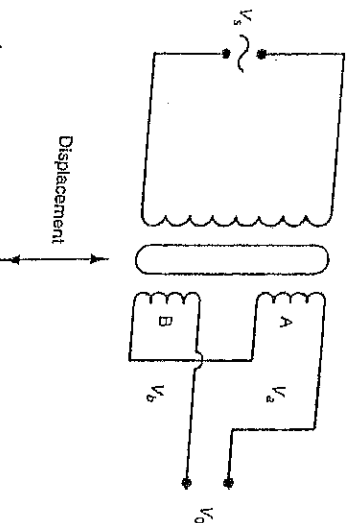


Fig 5: schematic diagram of an LVDT

Because of the series opposition mode of connection of secondary windings, $V_o = V_a - V_b$, and hence with the core in the central position, $V_o = 0$. Suppose now that the core is displaced upward (i.e., toward winding A) by distance x . If then $K_a = K_1$ and $K_b = K_2$, we have $V_o = (K_1 - K_2) \sin(\omega t - \phi)$.

If, alternatively, the core were displaced downward from the null position (i.e., toward winding B) by distance x , the values of K_a and K_b would then be $K_a = K_2$ and $K_b = K_1$, and we would have: $V_o = (K_2 - K_1) \sin(\omega t - \phi) = (K_1 - K_2) \sin(\omega t - \phi + \pi)$.

Thus, for equal magnitude displacements $+x$ and $-x$ of the core away from the central (null) position, the magnitude of the output voltage, V_o , is the same in both cases. The only information about the direction of movement of the core is contained in the phase of the output voltage, which differs between the two cases by 180° . If, therefore, measurements of core position on both sides of the null position are required, it is necessary to measure the phase as well as the magnitude of the output voltage. The relationship between the magnitude of the output voltage and the core position is approximately linear over a reasonable range of movement of the core on either side of the null position and is expressed using a constant of proportionality C as $V_o = Cx$.

The only moving part in an LVDT is the central iron core. As the core is only moving in the air gap between the windings, there is no friction or wear during operation. For this reason, the instrument is a very popular one for measuring linear displacements and has a quoted life expectancy of 200 years.

Potentiometer:

Potentiometers are very common devices used to measure displacement. A linear potentiometer is used for linear measurements and an angular potentiometer is used for angular measurements. The linear potentiometer is a device in which the resistance varies as a function of the position of a slider, shown below.

With the supply voltage (V_s), the output voltage (V_o) will vary between zero and the supply voltage.

For linear potentiometer, the output is a simple linear function of the slider position. That is

$$V_o = \frac{x}{L} V_s$$

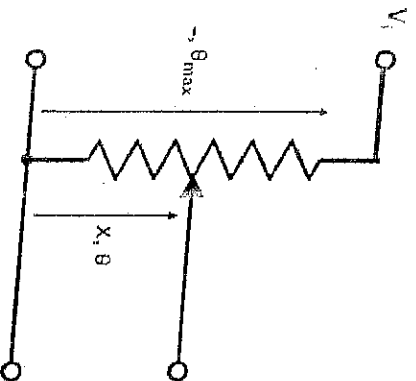
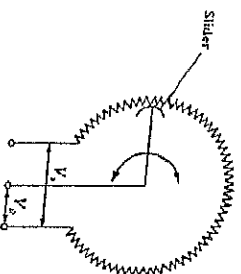
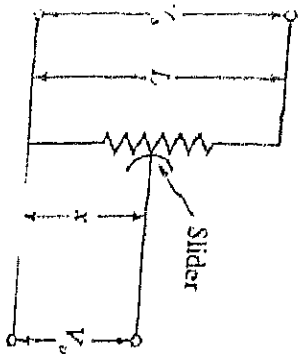


Fig 6: linear and angular potentiometer

$$V_o = \frac{\theta}{\theta_{max}} V_i$$

$$V_o = \frac{x}{L} V_i$$

It should be noted that the device measuring V_o must have a high impedance to maintain a linear response and avoid loading error.

Linear potentiometers can be used to measure displacements as small as 0.1 to 0.2 in. (2.5 to 5 mm) up to displacements of more than 1ft.

Measurement of force

Force can be said as any interaction of the object, such that it changes the objects motions or causes the change in velocity. In other words, if the force is unopposed, it will change the velocity of the body. The forces can be measured in two ways as follows:

1. Direct methods.
2. Indirect methods.

1. Direct Method: It involves a direct companion with known as gravitational force on the standard mass for Example Balancing mass.

Equal Arm-It is most simple force measuring system, its basic principle is based on the moment of comparison. This system consists of beam pivoted on a knife edge of fulcrum placed exactly at the center of the beam.

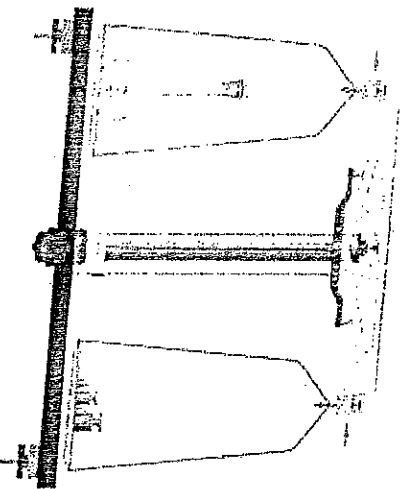


Fig. 7: equal arm balance
When the unknown force is applied at one end of the beam, the mass on the other end of the beam balances the unknown forces.

Analytical Balance

Analytical means analyzing or logical reason. With Figure shown there are two pivoted arms at point 'O' and two forces are held at the two points as W_1 and W_2 , taking W_2 as unknown force, The unbalance in the weight will be indicated by the arm in certain angular distance.

Unequal Arm balance

- The equal Arm is used to measure the heavier weight with the help of lighter weights. It uses two arms out of which one is called as load arm, and other is power arm.

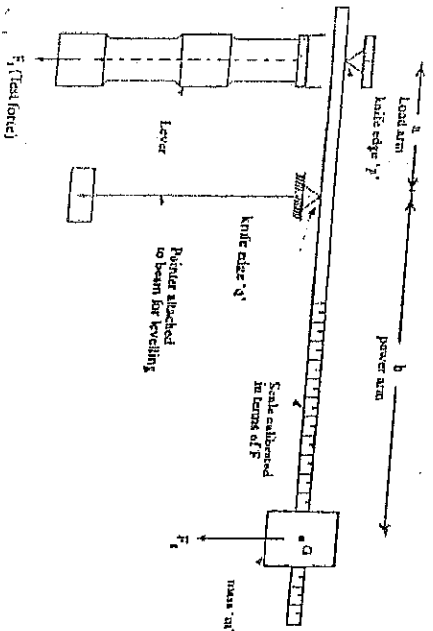


fig. 8: equal arm balance

- Load arm is where the load is applied and whereas the power arm is used to apply the counter weights to balance in equilibrium.
- Pendulum scale weight
- The input is direct force proportional to the weight. The input is transmitted with the suitable agency and applied to the load rod.
- As the load is applied the sectors rotate about the pivot moving the counter weights outwards.

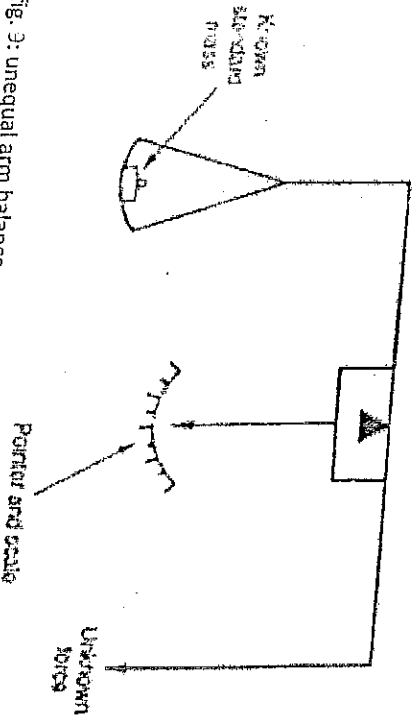


fig. 3: unequal arm balance
 2. Indirect methods: It involves the measurement of the effects of force on body such as acceleration of a body known mass subjected to force.

• Accelerometer-

A force will make a body to accelerate. By measuring the acceleration, the force may be determined from the equation $F = ma$

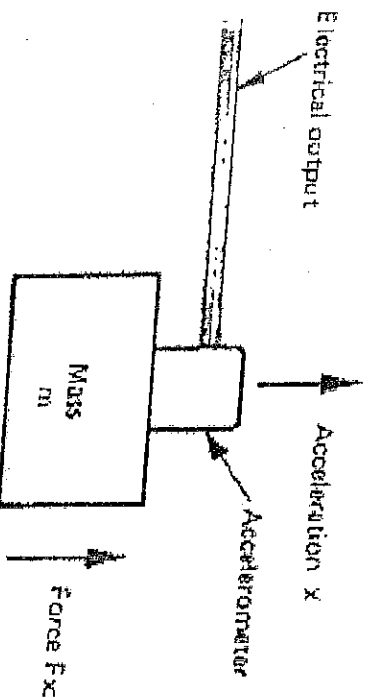


fig. 10: accelerometer
 Loadcell

Load cell is a electromechanical transducer that convert the loads/forces acting on it into an analog electrical signals. The major drawback is the need to minimize the friction.

Load cell provides accurate measurement of compressive and tensile load/forces.

Types of loadcell.

1. Hydraulic loadcell.

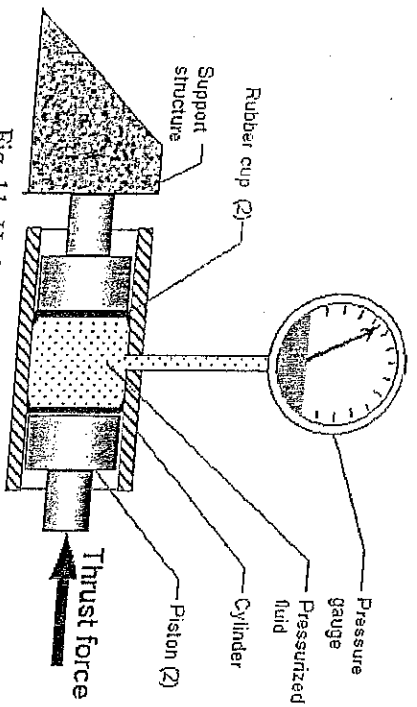


Fig. 11: Hydraulic Loadcell

2. Pneumatic loadcell.

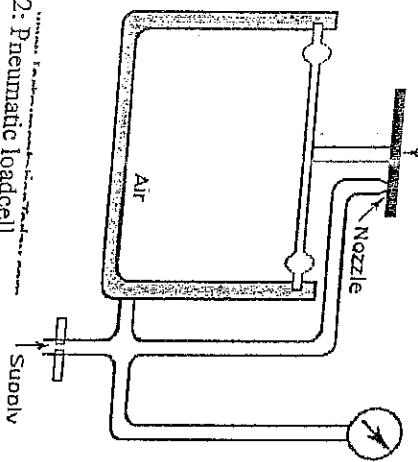


Fig. 12: Pneumatic loadcell

Elastic loaded members
It works with principle of finding the deflection or strain produced in the body to measure the force applied.

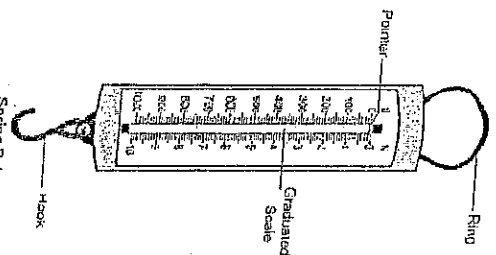


Fig. 13: a spring balance

The strain or deflection can be directly measured or in an indirect way by using secondary transducers by converting the displacement into another form of output which naturally electrical in nature.
Proving Rings
Proving ring is used to measure the displacement caused by the force.

The displacement transducer is placed on the both ends of the rings.
With LVDT transformer or Strain gauge the displacement is found out.

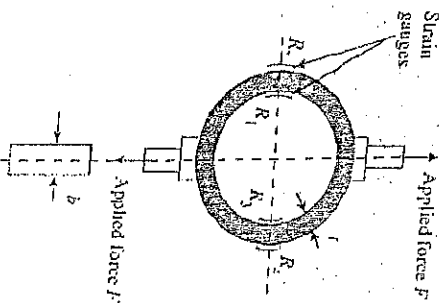


Fig. 14: proving rings

Measurement Of Torque

Torque is the just the rotational force through the distance, it is represented as moment vector force.

Types of Torque measurement devices
1. Prony brake. a Prony brake dynamometer comprises two wooden blocks that are mounted on either side of the fly wheel in diametrically opposite directions. The fly wheel is attached to the shaft

whose power needs to be determined. A lever arm is fixed to one block and the other arm is connected to an arrangement provided to tighten the rope. Tightening of the rope is performed in order to enhance the frictional resistance between the blocks and the flywheel. The torque exerted by the Prony brake is given by the following equation:

$$T = FL$$

Here, force F is measured by conventional force-measuring instruments such as load cells or balances.

The power dissipated in the brake is then calculated by the following equation:

$$P = \frac{2\pi NT}{60}$$

Or

$$P = \frac{2\pi nFL}{60} W$$

Here, P is the dissipated power in watts, L is the length of the lever arm in metres, N is the angular speed in revolution per minute, and F is the force in Newton.
Although a Prony brake is inexpensive, it is inherently unstable. It is difficult to adjust or to maintain specific loads.

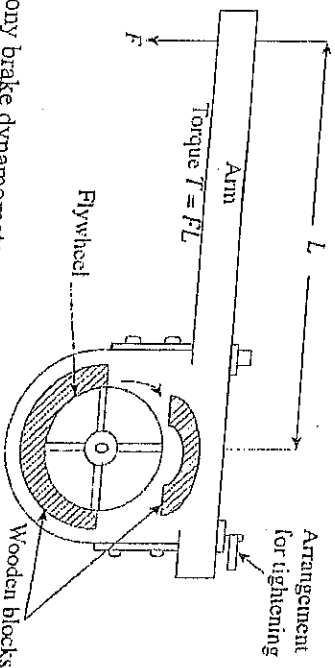


Fig.15: Prony brake dynamometer

2. Torque measurement using the strain gauge.
3. Torque measurement using torsional bar.

- i) Optical method.
- ii) Capacitive method.
- iii) Laser optic method.

Measurement Of Strain

What is Strain? Strain is the amount of deformation of a body due to an applied force. More specifically, strain is defined as the fractional change in length.

When a force is applied to a structure, the components of the structure change slightly in their dimensions and are said to be strained.

Devices to measure these small changes in dimensions are called *strain gauges*.

What devices can be used to measure strain?
Electrical Resistance Strain Gauge

The ideal sensor for the measurement of strain would
 Have good spatial resolution, implying that the sensor would

Be unaffected by changes in ambient conditions
 Have a high-frequency response for dynamic strain measurements.

A device that closely meets these characteristics is the *resistance strain gauge*.

A strain gauge, a device whose electrical resistance varies in proportion to the amount of strain in the device. The most widely used gauge is the bonded metallic strain gauge.

The electrical resistance strain gauge is an extremely common device used to measure strain in structures and also as a sensing element in a wide variety of transducers, including those used to

measure force, acceleration and pressure.

Electrical-resistance strain gauges and associated signal conditioners are simple, inexpensive and quite reliable.

To understand the function of a strain gauge, consider the measurement of strain in a simple structure shown below.

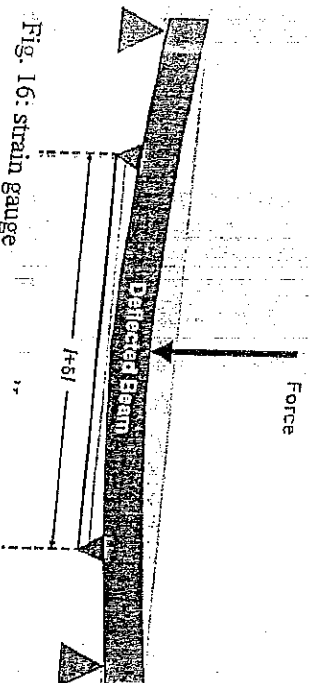


Fig. 16: strain gauge

Strain has units of inches per inch or millimeters per millimeter and hence it is dimensionless. In most structures the values of strain are usually very small. For example, low-strength steel will yield (take a permanent deform) at a strain about 0.0014.

Therefore, usually the strain is expressed in units of *micro strain* (μstrain).
 Thus $0.0014 \text{ strain} = 0.0014 \times 10^6 \mu\text{strain} = 1400 \mu\text{strain}$.

In the engineering design process, it is often necessary to determine the stresses in a structure experimentally to determine if the structure is sound.

It is difficult to measure the stress directly, but a strain gage can be used to measure the strain, and then the stress can be determined using the Hooke's law. That is

$$\sigma = E \epsilon$$

where, σ is the normal stress and E is the modulus of elasticity (also called young's modulus) which is a material property. For a wire to work as a strain gage, the relationship between the change in resistance and the strain must be known.

The resistance of a wire is given by

$$R = \frac{\rho L}{A}$$

where, R is the resistance, ρ is the resistivity of wire which is a function of the wire material, L is the length of wire, and A is the cross-sectional area of the wire. Taking logarithms of both sides, separating the terms and differentiating each term, we get

$$\frac{dR}{R} = \frac{d\rho}{\rho} + \frac{dL}{L} - \frac{dA}{A}$$

The above equation relates a small change in resistance to changes in resistivity, length and cross-sectional area.

The term dL/L is the *axial strain*, ϵ_a .

The term dA/A can be evaluated from the equation of the cross-sectional area $A = \pi D^2/4$.

Taking the logarithm and differentiating the above equation we get

$$\frac{dA}{A} = 2 \frac{dD}{D}$$

The term dD/D is known as the *transverse strain*, ϵ_t .

Solid mechanics provides the following relationship between the axial and Transverse strain

where, ν is known as Poisson's ratio and it is the property of material. The negative sign indicates that as the wire becomes longer, the transverse dimension decreases.

Combining the above equations, we get

$$\frac{dR}{R} = \frac{d\rho}{\rho} + \epsilon_a (1 + 2\nu)$$

The above equation shows the relationship between the change in resistance of the wire, strain, and the change in resistivity of the wire.

The strain gage factor, S , is defined as

$$S = \frac{dR/R}{\epsilon_a}$$

Combining the above two equations we get

$$S = 1 + 2\nu + \frac{d\rho/\rho}{\epsilon_a}$$

If the temperature is held constant, the change in resistivity is proportional to the strain.

• The strain gage factor is approximately constant, although it is sensitive to the temperature change.

• In addition to the strain gages constructed in the form of straight wires, another common type of strain gauges are constructed by etching them from thin foil metal sheets that are bounded to a plastic backing, as shown below.

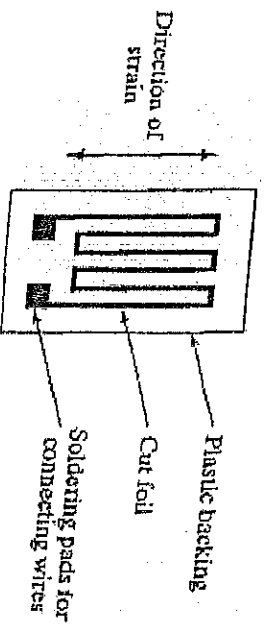


Fig. 17: foil strain gauge

This backing is glued to the structure whose strain needs to be measured.

The dimensions of strain gages vary. They can be as small as 200 μm .

Strains as high as 200,000 $\mu\epsilon$ can be measured.

Strain gages can also be constructed from semiconductor materials.

The semiconductor strain gages are commonly used as sensing elements in pressure and acceleration transducers. However, they cannot measure very high strain.

If a structure is loaded in a single direction, there exists a transverse strain but no transverse stress.

This strain can be measured as

$$\epsilon_t = -\nu \epsilon_a$$

This effect is included when manufacturers determine strain gage factors.

In many situations, the surface of a structure is stressed simultaneously in more than one direction, leading to the so-called *biaxial stress*.

In biaxial stress there is a transverse strain that results from the transverse stress.

This transverse strain affects the strain gage output and be described with a transverse gauge factor, S_t , defined as

$$S_t = \frac{dR/R}{\epsilon_t}$$

The transverse sensitivity effects are usually neglected in the strain measurements.

To define the strain on a surface, it is necessary to specify two orthogonal linear strains ϵ_x and ϵ_y , and a third strain called the shear strain, γ_{xy} , the change in angle between two originally orthogonal lines when the solid is strained.

• These strains can be determined by three suitably placed strain gages in an arrangement called a *strain rosette*.

Two common arrangements of the three strain gages in an arrangement called a *Rectangular rosette*.

- Equiangular rosette
- In rectangular rosette, the gages are placed at angles of 0, 45 and 90 degrees.
- In equiangular rosette, the gages are arranged at 0, 60 and 120 degrees.

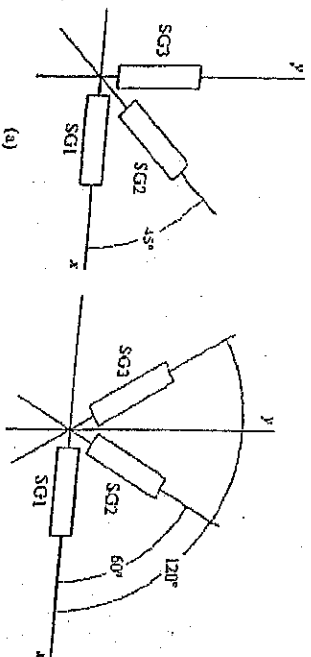


Fig. 18: strain gauge rosettes (a) Rectangular (b) Equiangular
Measurement Of Angular Velocity

Speed is a rate variable defined as the time- rate of motion. Common forms and units of speed measurement include: linear speed expressed in meters per second (m/s), and the angular speed of a rotating machine usually expressed in radians per second (rad/s) or revolutions per minute (rpm). Measurement of rotational speed has acquired prominence compared to the measurement of linear speed. Angular measurements are made with a device called tachometer. Tachometers may be broadly classified into two categories:

Mechanical tachometers and

Electrical tachometers.

The revolution counter, sometimes called a speed counter, consists of a worm gear which is also the shaft attachment and is driven by the speed source.

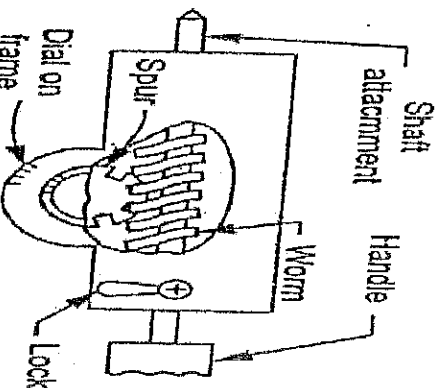


Fig. 19: Revolution counter

The worm drives the spur gear which in turn actuates the pointer on a calibrated dial. A properly designed and manufactured revolution counter would give a satisfactory speed measure up to 2000-3000 rpm.

Tachoscope

The difficulty of starting a counter and a watch at exactly the same time led to the development of tachoscope, which consists of a revolution counter incorporating a built-in timing device. The two components are integrally mounted, and start simultaneously when the contact point is pressed against the rotating shaft. The instrument runs until the contact point is disengaged from the shaft.

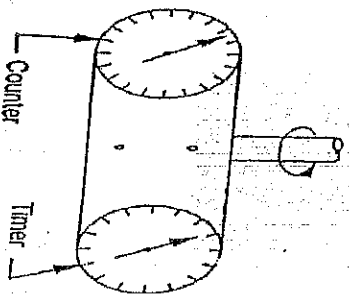


Fig. 20: a tachoscope

The rotational speed is computed from the readings of the counter and timer. Tachoscopes have been used to measure speeds up to 5000 rpm.

Hand Speed Indicator

The indicator has an integral stopwatch and counter with automatic disconnect. The indicator operates when brought in contact with the shaft, but the counter does not function until the start and wind button is pressed to start the watch and engage the automatic clutch. Depressing of the starting button also serves to wind the starting watch. After a fixed time-interval (usually 3 or 6 seconds), the revolution counter automatically gets disengaged.

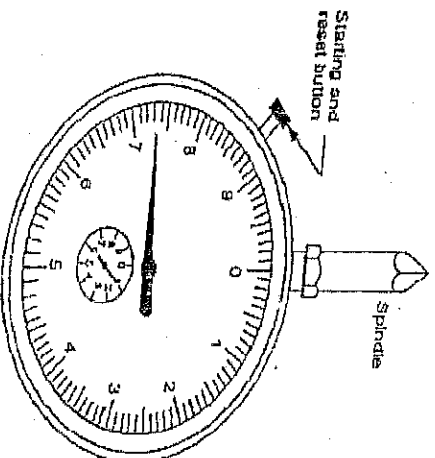


Fig. 21: hand speed indicator

Slipping Clutch Tachometer

The rotating shaft drives an indicating shaft through a slipping clutch. A pointer attached to the indicator shaft moves over a calibrated scale against the torque of a spring. The pointer position gives a measure of the shaft speed.

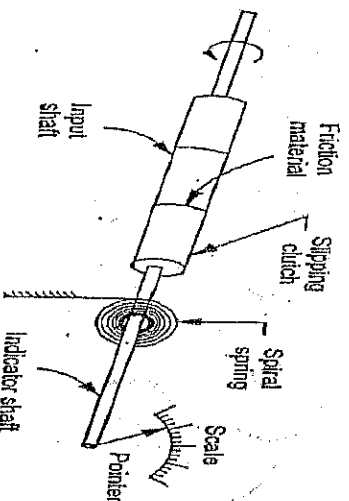


Fig. 22: slipping clutch tachometer

Centrifugal Force Tachometer

The device operates on the principle that centrifugal force is proportional to the speed of rotation. Two fly balls (small weights) are arranged about a central spindle. Centrifugal force developed by these rotating balls works to compress the spring as a function of rotational speed. A grooved collar or sleeve attached to the free end of the spring then slides on the spindle and its position can be calibrated in terms of the shaft speed.

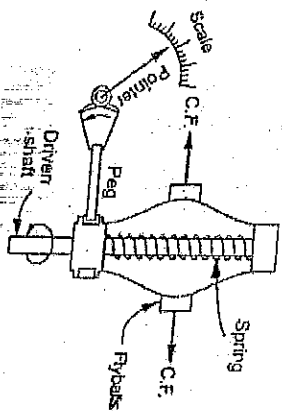


Fig. 23: centrifugal tachometer
Vibrating Tachometer

Tachometers of the vibrating reed type utilize the fact that speed and vibration in a body are interrelated. The instrument consists of a set of vertical reeds, each having its own natural frequency of vibration. The reeds are lined up in order of their natural frequency and are fastened to a base plate at one end, with the other end free to vibrate.

Electrical Tachometers

An electrical tachometer depends for its indications upon an electrical signal generated in proportion to the rotational speed of the shaft. Depending on the type of the transducer, electrical tachometers have been constructed in a variety of different designs.

Drag Cup Tachometer

In an eddy current or drag type tachometer, the test shaft rotates a permanent magnet and this induces eddy currents in a drag cup or disc held close to the magnet. The eddy currents produce a torque which rotates the cup against the torque of a spiral spring. The disc turns in the direction of the rotating magnetic field until the torque developed equals that of the spring. A pointer attached to the cup indicates the rotational speed on a calibrated scale.

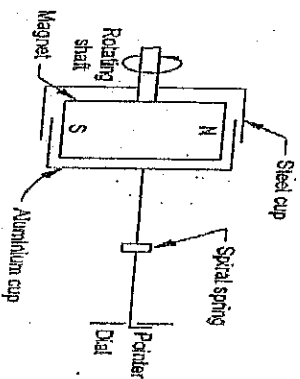


Fig. 24: drag cup tachometer

Tachogenerators

These tachometers employ small magnet typed d.c. or a.c. generators which translate the rotational speeds into d.c. or a.c. voltage signal. The operating principle of such tachometers is illustrated in the Fig. below

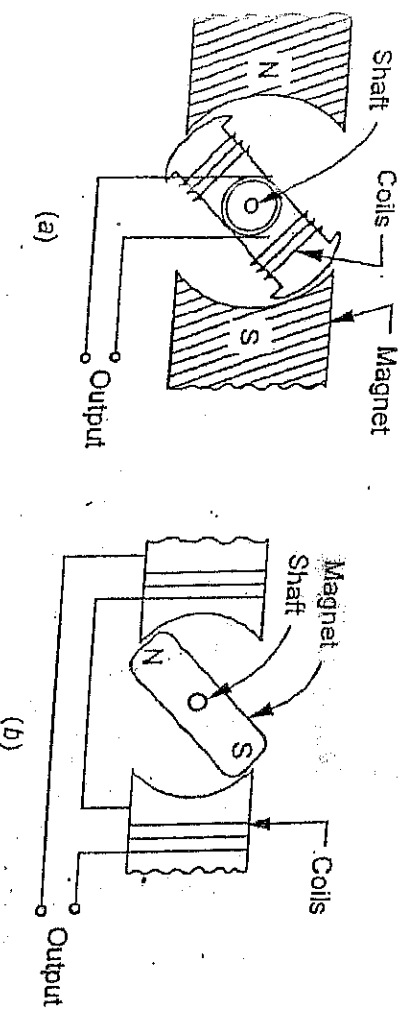


Fig. 25: tachogenerators (a) dc tachometer and (b) ac tachometer

Relative perpendicular motion between a magnetic field and conductor results in voltage generation in the conductor.

D.C. tachometer generator: This is an accurately made dc. Generator with a permanent magnet of horse-shoe type.

A.C. tachometer generator: The unit embodies a stator surrounding a rotating permanent magnet

Stroboscope

The stroboscope utilizes the phenomenon of vision when an object is viewed intermittently. The human sense of vision is so slow to react to light stimuli that it is unable to separate two different light impulses reaching the eye within a very short Period of time (less than 0.1 second). A succession of impulses following one another at brief intervals are observed by the eye as a continuous unbroken sequence.

A mechanical disk type stroboscope consists essentially of a whirling disk attached to motor whose speed can be varied and measured. A reference mark on the rotating shaft on the shaft appears to be stationary.

Temperature Measuring Systems

Temperature: the temperature of a substance is a measure of the hotness, or the coldness of that substance. It is the thermal state of the body which determines whether it will give heat to, or receive heat from other bodies. The term temperature and heat are closely related. Temperature may be defined as "degree of heat" but heat is usually taken to mean "quantity of heat". Temperature and heat are related quantitatively since heat flows, of its own accord, from a body at higher temperature to a body at a lower temperature. It is therefore important to remember that in temperature measurement, two bodies in intimate contact are at the same temperature only if there is no heat flow between them.

Temperature scales: Temperature scales are based upon some recognized fixed points. At least two fixed points are required which are constant in temperature and can be easily reproduced such as: the lower fixed point, or ice-point
the upper fixed point, or steam-point

The lower fixed point, or ice point, is the temperature of ice, prepared from distilled water, when melting under a pressure of 760 mm hg. The upper fixed point is the temperature of steam from pure distilled water: boiling under a pressure of 760 mm hg. The boiling point of water varies greatly with applied pressure. Thus, it is important to note the pressure at which the water is boiling.

The temperature difference between the ice point and the steam point is known as the "fundamental interval". In other to graduate a thermometer between these fixed points, the temperature between these points is divided into a number of equal parts.

Convert Reline to Gears.

$$D_K = D_C + 2.73-5$$

Convert Rankin to Facewidth

$$D_R = 4F + 1.91.9$$

Convert Gears to Facewidth

$$D_C = 4F - 32$$
$$\frac{100}{150}$$

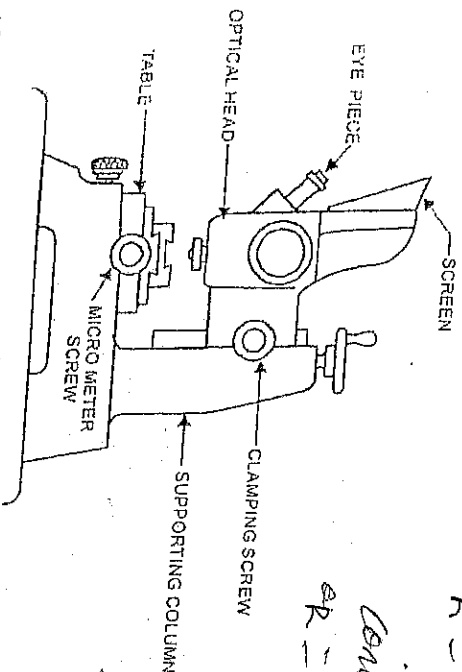


Fig. 46: tool makers microscope

GEAR MEASUREMENT ✓

Introduction

Gear is a mechanical drive which transmits power through toothed wheel. In this gear drive, the driving wheel is in direct contact with driven wheel. The accuracy of gearing is the very important factor when gears are manufactured. The transmission efficiency is almost 99 in gears. So, it is very important to test and measure the gears precisely. For proper inspection of gear, it is very important to concentrate on the raw materials, which are used to manufacture the gears, also gear blanks should be tested for machining the blanks, heat treatment and the finishing of teeth. The most commonly used forms of gear teeth are:

1. Involute

2. Cycloidal

The involute gears also called as straight tooth or spur gears. The cycloidal gears are used in heavy and impact loads. The involute rack has straight teeth. The involute pressure angle is either 20° or 14.5° .

Types of gears ✓

1. **Spur gear:** cylindrical gear whose tooth traces is straight line. These are used for transmitting power between parallel shafts.
2. **Spiral gear:** The tooth of the gear traces curved lines.
3. **Helical gears:** These gears used to transmit the power between parallel shafts as well as non-parallel and nonintersecting shafts. It is a cylindrical gear whose tooth traces is straight line.
4. **Bevel gears:** The tooth traces are straight-line generators of cone. The teeth are cut on the conical surface. It is used to connect the shafts at right angles.
5. **Worm and Worm wheel:** It is used to connect the shafts whose axes are non-parallel and non-intersecting.
6. **Rack and Pinion:** Rack gears are straight spur gears with infinite radius.

Gear terminology ✓

1. **Tooth profile:** It is the shape of any side of gear tooth in its cross section.
2. **Base circle:** It is the circle of gear from which the involute profile is derived. Base circle diameter = Pitch circle diameter $\times$ Cosine of pressure angle of gear
3. **Pitch circle diameter (PCD):** The diameter of a circle which will produce the same motion as the toothed gear wheel.
4. **Pitch circle:** It is the imaginary circle of gear that rolls without slipping over the circle of its mating gear.
5. **Addendum circle:** The circle coincides with the crests (or) tops of teeth.
6. **Dedendum circle (or) Root circle:** This circle coincides with the roots (or) bottom on teeth.
7. **Pressure angle (α):** It is the angle making by the line of action with the common tangent to the pitch circles of mating gears.
8. **Module (m):** It is the ratio of pitch circle diameter to the total number of teeth. Where, $d =$ Pitch circle diameter, $n =$ Number of teeth.
9. **Circular pitch:** It is the distance along the pitch circle between corresponding points of adjacent teeth.
10. **Addendum:** Radial distance between tip circle and pitch circle. Addendum value = 1 module.
11. **Dedendum:** Radial distance between pitch circle and root circle. Dedendum value = 1.25 module.

12. **Clearance (C):** Amount of distance made by the tip of one gear with the root of mating gear.
Clearance = Difference between Dedendum and addendum values.
13. **Blank diameter:** The diameter of the blank from which gear is cut. Blank diameter = $PCD + 2m$
14. **Face:** Part of the tooth in the axial plane lying between tip circle and pitch circle.
15. **Flank:** Part of the tooth lying between pitch circle and root circle.
16. **Top land:** Top surface of a tooth.
17. **Lead angle:** The angle between the tangent to the helix and plane perpendicular to the axis of cylinder.
18. **Backlash:** The difference between the tooth thickness and the space into which it meshes.

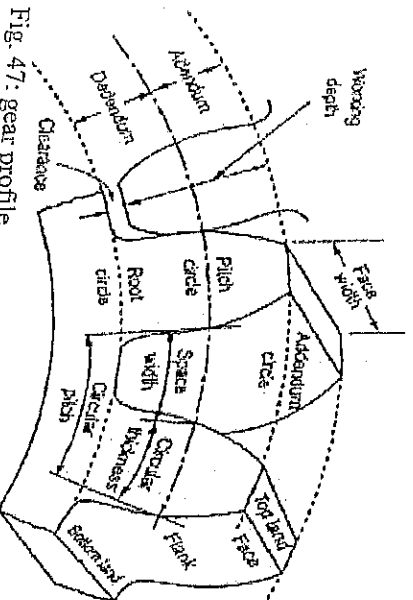


Fig. 47: gear profile

The Inspection of the gears consists of determine the following elements in which manufacturing error may be present.

1. Runout.
2. Pitch
3. Profile
4. Load
5. Back lash
6. Tooth thickness
7. Concentricity
8. Alignment

Surface Finish Measurement Introduction

When we are producing components by various methods of manufacturing process it is not possible to produce perfectly smooth surface and some irregularities are formed. These irregularities are causing some serious difficulties in using the components. So, it is very important to correct the surfaces before use. The factors which are affecting surface roughness are:

1. Work piece material
 2. Vibrations
 3. Machining type
 4. Tool and fixtures
- The geometrical irregularities can be classified as

1. First order
2. Second order
- 3.3 Third order
4. Fourth order

1. **First order irregularities:** These are caused by lack of straightness of guide ways on which tool must move.
2. **Second order irregularities:** These are caused by vibrations
3. **Third order irregularities:** These are caused by vibrations
4. **Fourth order irregularities:** These are caused by machining equipment.

Elements of surface texture

1. **Profile:** - Contour of any section through a surface.
2. **Lay:** - Direction of the 'predominate surface pattern'
3. **Flaws:** - Surface irregularities or imperfection, which occur at infrequent intervals.
4. **Actual surface:** - Surface of a part which is actually obtained.
5. **Roughness:** - Finely spaced irregularities. It is also called primary texture.

Effective diameter measurement is carried out by following methods.

1. One wire,
2. Two wires, or
3. Three wires method.
4. Micrometer method.

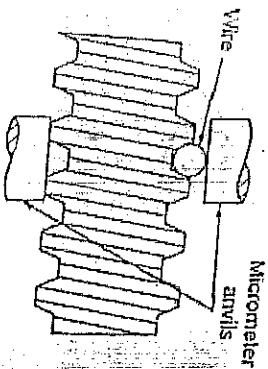


Fig. 42: one wire method

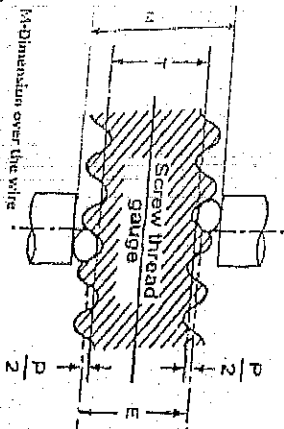


Fig. 43: two wire method

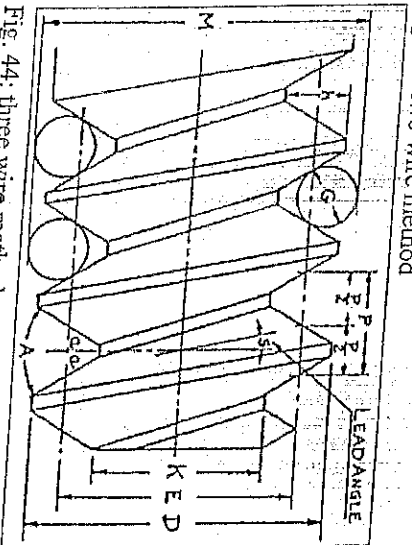


Fig. 44: three wire method

- The most commonly used methods for measuring the pitch are
1. Pitch measuring machine
 2. Tool maker's microscope
 3. Screw pitch gauge

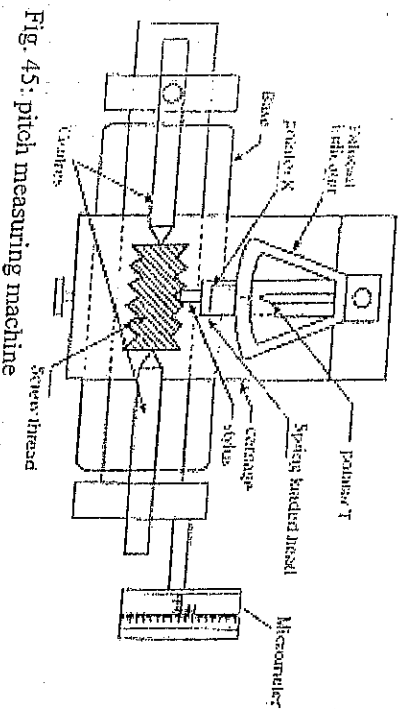


Fig. 45: pitch measuring machine

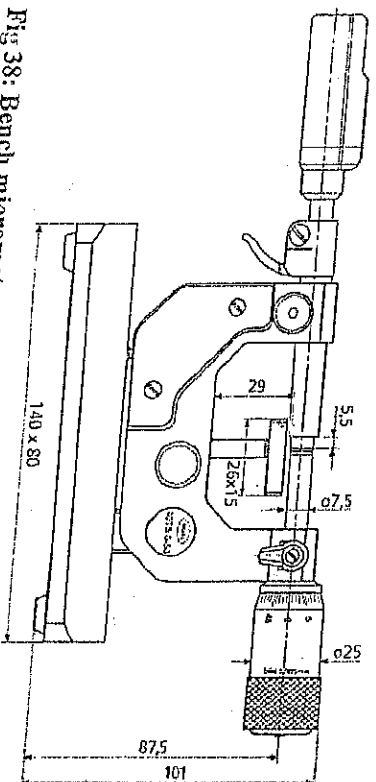


Fig. 38: Bench micrometer

2. Measurement of Minor diameter floating carriage diameter measuring machine and small V pieces.

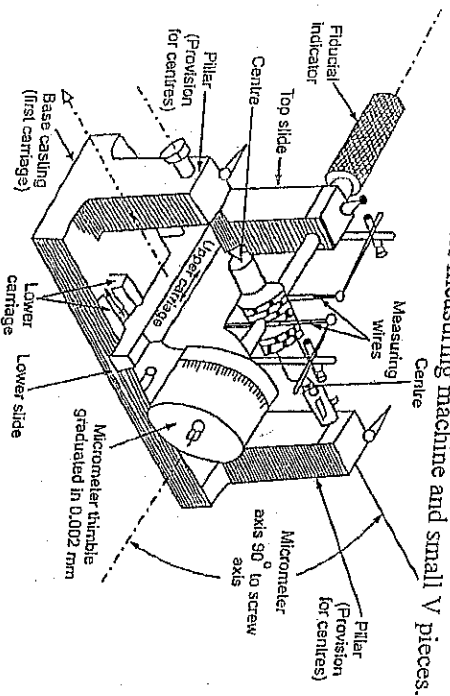


Fig. 39: floating carriage diameter measuring machine

The Minor diameter of Internal threads are measured by

1. Using taper parallels
2. Using Rollers.

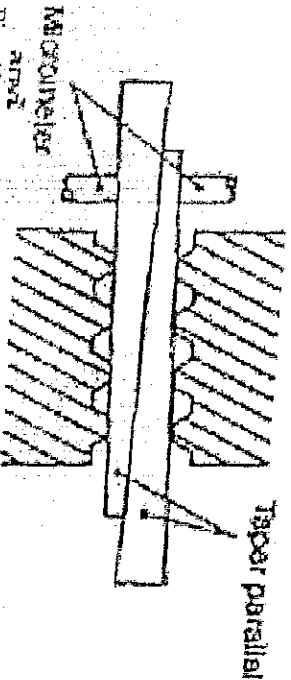


Fig. 40: taper parallel

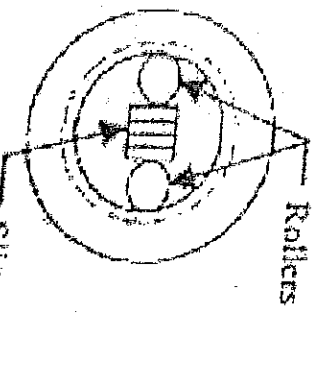


Fig. 41: rollers

3. Measurement of effective diameter

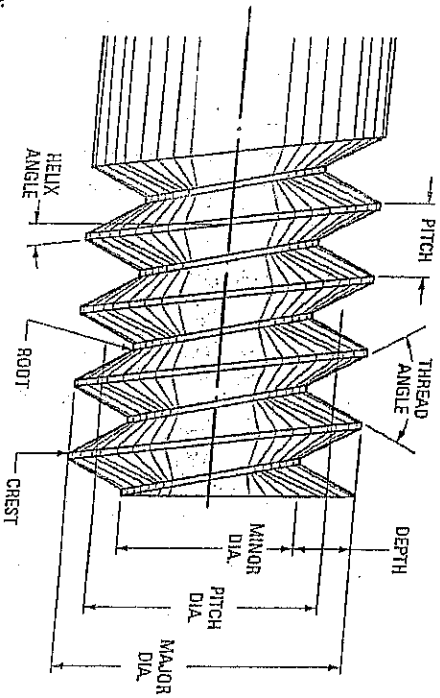


Fig. 37: Screw Thread Terminology

Screw Thread Terminology

Pitch: It is the distance measured parallel to the screw threads axis between the corresponding points on two adjacent threads in the same axial plane. The basic pitch is equal to the corresponding divided by the number of thread starts.

Minor diameter: It is the diameter of an imaginary co-axial cylinder which touches the roots of external threads.

Major diameter: It is the diameter of an imaginary co-axial cylinder which touches the roots of of an external thread and the root of an internal thread.

Pitch diameter: It is the diameter at which the thread space and width are equal to half of the screw thread

Helix angle: It is the angle made by the helix of the thread at the pitch line with the axis. The angle is measured in an axial plane.

Flank angle: It is the angle between the flank and a line normal to the axis passing through the apex of the thread.

Height of thread: It is the distance measured radially between the major and minor diameters respectively

Addendum: Radial distance between the major and pitch cylinders for external threads

Radial distance between the minor and pitch cylinder for external thread.

Dedendum: It is the radial distance between the pitch and minor cylinders for internal thread.

Also, radial distance between the major and pitch cylinders for internal thread.

Measurement of various elements of Thread

To find out the accuracy of a screw thread it will be necessary to measure the following:

1. Major diameter.
2. Minor diameter.
3. Effective or Pitch diameter.
4. Pitch
5. Thread angle and form

1. Measurement of major diameter:

- The instruments which are used to find the major diameter are by
- Ordinary micrometer
 - Bench micrometer.

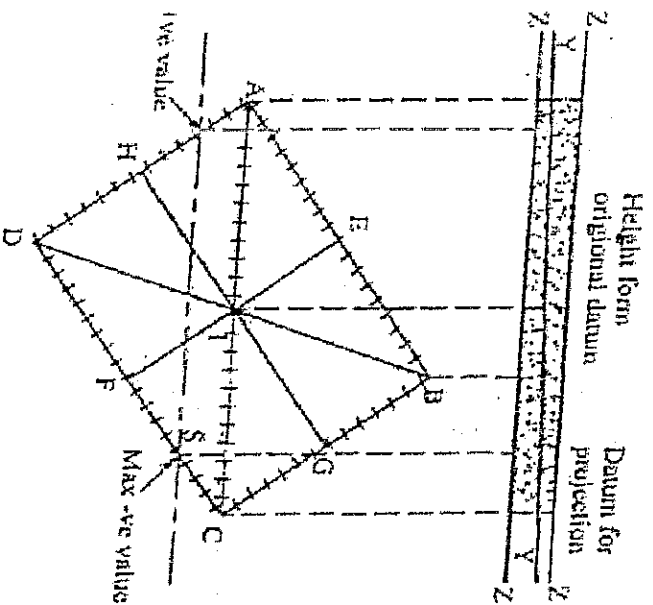


Fig. 35: Flatness Testing

Thread measurement

Threads are of prime importance; they are used as fasteners. It is a helical groove, used to transmit force and motion. In plain shaft, the hole assembly, the object of dimensional control is to ensure certain consistency of fit. The performance of screw threads during their assembly with nut depends upon a number of parameters such as the condition of the machine tool used for screw cutting, work material and tool.

Screw threads are used to transmit the power and motion, and also used to fasten two components with the help of nuts, bolts and studs. There is a large variety of screw threads varying in their form, by included angle, head angle, helix angle etc. The screw threads are mainly classified into 1) External thread 2) Internal thread.

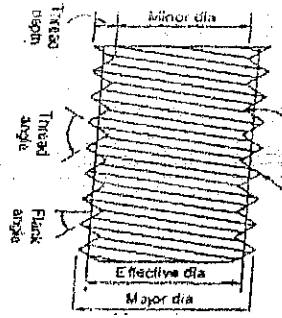
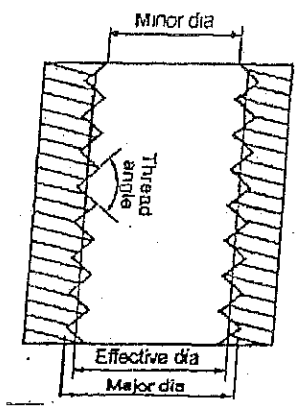


Fig. 36: a) External Thread



b) Internal thread

A block resting on two legs or feet is placed on the surface under test. Block carries a plane vertical reflector mounted on its extreme left side in such a way that, face of reflector is facing auto-collimator.

- i Plane reflector and auto-collimator are arranged in such a way that,
 - (i) image of cross wires of the collimator appears very near to the centre of eyepiece, and
 - ii (ii) linear movement of reflector over the entire length of surface under test is completed.

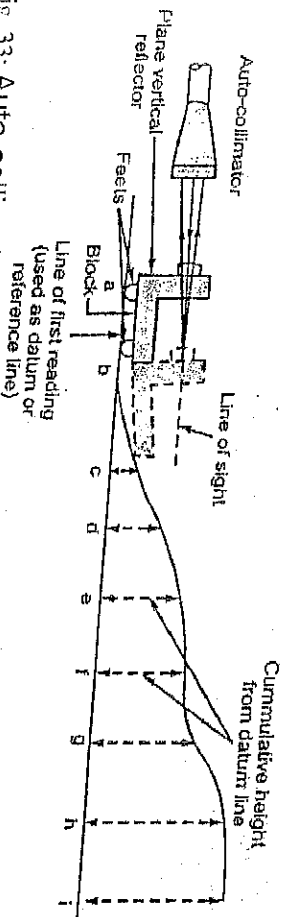


Fig. 33: Auto-collimator checking straightness

Now, the reflector is moved towards the other end of surface in steps equal to the centre distance between the two legs or feet. During this movement, the tilting of reflector is noted down in seconds from the eyepiece.

1 second of arc = 0.000006 mm/mm

Now, the reflector is set at first position a-b (perfectly flat and straight) and first micrometer reading is noted down. This line is labelled as 'a-b' is treated as datum line or reference line. Successive readings at positions b-c, c-d, d-e and so on, are taken, till the plane reflector completes its linear movement over the entire length of surface under test.

Measurement of Flatness

It is defined as minimum distance between two planes within which all the points on a surface lie. A surface along which all the points lie along single plane is called as perfectly flat surface. Flatness testing is possible by comparing the surface with an accurate surface.

that the departure from flatness is the minimum separation of a pair of parallel planes which will contain all points on the Surface. The figure which shows that a surface can be considered to be straight and they lie in the same plane. In the case of rectangular table the lines are straight and parallel to the sides of the rectangle in both the perpendicular direction. Even it is not flat, but concave and convex along two diagonals. For verification, it is essential to measure the straightness of diagonals in addition to the lines parallel to the sides.

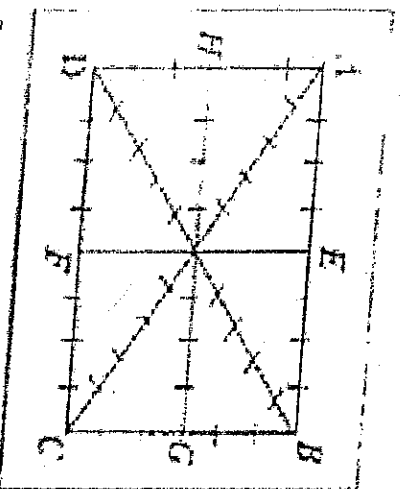


Fig. 34: flatness testing procedure

The fig. 34 shows the flatness testing procedure.

- (i) Carry out the straightness test and tabulate the reading up to the cumulative error column.
- (ii) Ends of lines AB, AD and BD are corrected to zero and thus the height of the points A, B and D are zero.

PRINCIPLES OF FORM MEASUREMENT

Form measurement includes

- Screw thread measurement
- Gear measurement
- Radius measurement
- Surface Finish measurement
- Straightness measurement
- Flatness and roundness measurements

Principles and Methods of straightness

"A line is said to be straight over a given length, if variation in distance of all points lying on line from two planes perpendicular to each other and parallel to general direction of line, remains within the specified tolerance limits".

The reference planes are so chosen that, their intersection is parallel to the straight line joining two points, which are located on the line to be tested and close to ends of the length under measurement. Tolerance on straightness of a line is defined as, "the maximum deviation in relation to the reference straight line going to the two extremities or ends of line under examination".

Test for Straightness by Using Spirit Level and Auto-collimator

Tests for straightness can be carried out by using spirit level or autocollimator.

The above instruments determine the straightness of any surface by measuring the relative angular position of various adjacent sections of surface to be tested.

For this purpose, initially a straight line is drawn on the surface to be tested.

divided into number of equidistant sections.

If spirit level is used, then length of each section should be equal to length of base of spirit level.

If auto-collimator is used, then length of each section should be equal to length of base of plane reflector.

Generally, the bases of spirit level block or reflector are fitted with two legs (or feet), such that

- (i) Feet or legs have line contact with the surface under test, and
- (ii) Entire surface of base does not touch the surface under test

This ensures that, angular deviation obtained is between the two specified points.

(i) Spirit Level:

The block of spirit level is moved linearly on the surface to be tested, in number of steps. Every step chosen is equal to the pitch distance between centre lines of two feet. When block of spirit level is kept on a perfectly flat surface, we observe that, vapour bubble is resting at the middle position. But, when the block of spirit level is moved over surface test, then this vapour bubble moves away from the middle position due to irregularities in straightness of surface. This variation in the spirit level is measured, which gives the angular variation in the direction of block.

Angular variation can be correlated in terms of the difference of height between two points by knowing the least count of level and length of the base.

Limitation of spirit level: It can be used to find out variation in straightness of horizontal surface only.

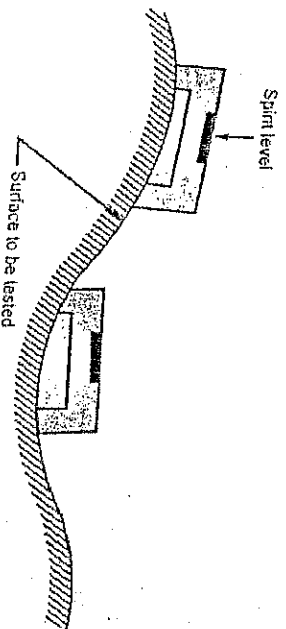


Fig. 32: Test for Straightness by Using Spirit Level

(ii) Auto-collimator:

Auto-collimator can be used to measure the straightness of surface in any plane. Auto-collimator is placed at a distance of 0.5 to 0.75 metre from the surface to be tested. It is held in desired position on any rigid support, which is independent of the surface to be tested.

A parallel beam from auto-collimator is projected along the length of surface to be tested.

contracts and it forces the bourdon tube to coil more tightly. The movement of the bourdon tube may be used to drive a pointer for indicating temperature.

Fig 30: filled system thermometer

Basically, there are four types of filled system thermometers:

Gas filled thermometers (already discussed)

Liquid filled thermometers

Mercury filled thermometers

Vapour pressure thermometers

Liquid-Filled Thermometers: these thermometers are completely filled with liquids (other than mercury) and operate on the principle of liquid expansion with increase in temperature. The filling is usually an inert hydrocarbon such as xylene (C_8H_{10}) which has a co-efficient of expansion six times that of mercury and makes smaller bulb possible. Other liquids (even water) are sometimes used. The criterion is that the pressure inside the system must be greater than the vapour pressure of the liquid to prevent bubbles of vapour from forming. Also, the liquid should not be allowed to solidify even in storage; otherwise, the calibration may be affected.

Vapour Pressure Thermometers: in this system the bulb is partially filled with liquid, while the capillary tube and the bourdon are filled with vapour. In this system some of the liquid vaporizes during operation.

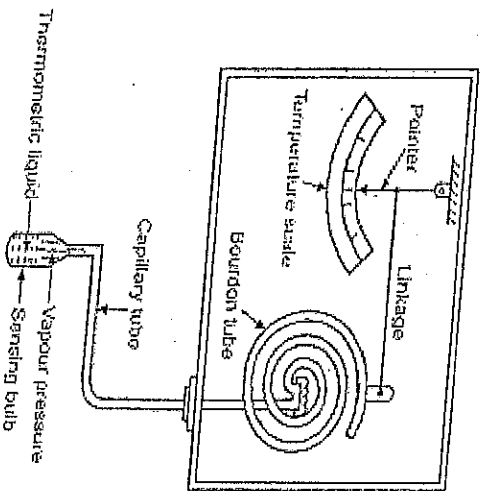


Fig 31: vapour pressure thermometer.

A vapour pressure thermometer is shown in fig 31. The liquid in the system boils and vaporizes during operation which creates a gas or vapour inside the capillary and bourdon tube. The liquid inside the bulb continues to boil until the pressure in the system equals the vapour pressure of the boiling liquid. At this point the liquid stops boiling unless its temperature increases. When the temperature of the substance surrounding the bulb drops, the liquid and the vapour inside the bulb also cool, causing some of the vapour to condense. As the vapour condenses, the pressure inside the system decreases. Due to this change in pressure, the bourdon tube uncoils as the pressure increases and coils as it decreases. This movement of the bourdon tube may be connected to a pointer, or to a pen on a strip chart recorder, or to a transmitter to indicate temperature.

and the glass capillary by one of stainless steel. Mercury is used as liquid in the system. As mercury in the system is not visible, a bourdon is used to measure the change in its volume. The bourdon, the capillary tube, and the bulb are completely filled with mercury, usually at higher pressure.

Fig 29: mercury in steel thermometer.

When the temperature to be measured rises, the mercury in the bulb expands more than the bulb so that some mercury is driven through the capillary tube to the bourdon tube. As the temperature continues to rise, increasing amount of mercury will be driven into the bourdon tube, causing it to bend. One end of the bourdon tube is fixed, while the motion of the other end is communicated to the pointer which moves on a calibrated temperature scale.

Generally, mercury is used as a liquid. But it has its limitations, particularly at the lower end of the temperature scale. For this and other reasons, other liquids are also used in place of mercury; these include xylene, alcohol, ether etc.

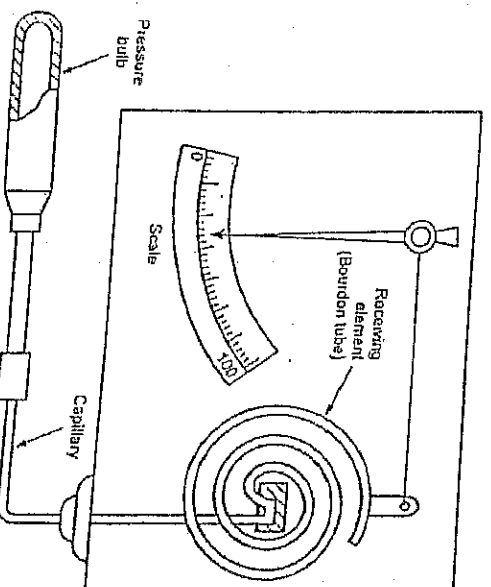
Gas Thermometers: the operation of the gas thermometer is based on the ideal gas law which states that the volume of a gas increases with temperature, if the pressure is maintained constant; and the pressure increases with temperature if the volume is maintained constant. Therefore, if a certain volume of inert gas is contained in a bulb, Capillary and bourdon tube, and must of the gas in the bulb, then the pressure indicated by the bourdon tube may be calibrated in terms of temperature of the bulb.

Nitrogen is the favorite fill for the gas filled thermometer because it is almost inert and inexpensive. It does react somewhat with the steel at temperatures exceeding 427°C , and it does act less like a perfect gas at extremely low temperatures. Under these conditions, helium should be used.

An advantage of the gas thermometer is that the gas in the bulb has a lower thermal capacity than a similar quantity of liquid, so that the response to temperature changes will be more rapid than for a liquid-filled system with a bulb the same size and shape. It has a temperature range of -200 to 1500°C depending on the gas used.

Filled System Thermometers:

These consist of a bourdon tube, a capillary tube and a thermometer bulb all interconnected as shown in fig 1.6. The entire system is sealed after filling with the appropriate liquid (known as filling liquid) at a pressure at the normal ambient temperature. The common liquids used are mercury, ethyl alcohol, xylene and toluene. When in use the thermometer bulb is inserted inside the substance to be measured. This causes the filling liquid inside the bulb to heat or cool until its temperature matches that of the substance. This change in temperature causes the filling liquid to expand or contract and thus the bourdon tube moves. With increase in temperature the liquid expands and this forces the bourdon tube to uncoil. With decrease in temperature the liquid



Liquid-In-Glass Thermometers: the liquid in glass thermometer is one of the simplest temperature measuring devices, widely used in both laboratory and industry. Its operation is based on the fact that liquids expand as the temperature rises. In this type of thermometer, the expansion causes the liquid to rise in the tube, indicating the temperature. The simplest form of liquid in glass thermometer is shown in fig 1.4. It consists of a small-bore glass tube with a thin-walled glass bulb at its lower end containing almost all the liquid (usually mercury). As heat is transferred into the mercury, the mercury expands, pushing the column of mercury higher in the capillary above which indicates the temperature.

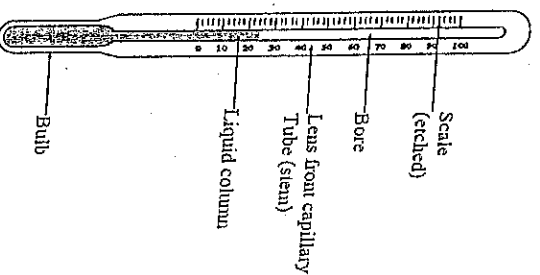
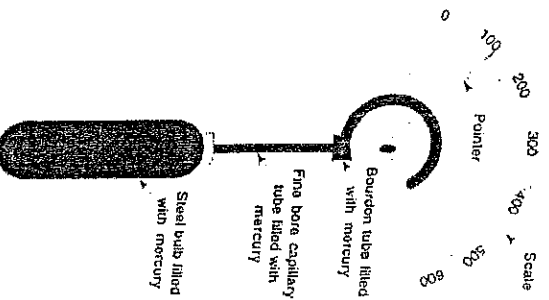


Fig 28: liquid in glass thermometer.

Liquid-in-glass thermometer is commonly used for temperature range of -120 to 320°C . But when mercury is used as liquid, it freezes at -39°C . Thus, for measuring very low temperature, alcohol is used as liquid. This type of thermometer cannot be used to measure temperatures higher than 600°C , because it permanent changes in the volume of the bulb, thus destroying the accuracy of the instrument.

The industrial mercury-in-glass thermometer is used in applications such as open tanks containing liquids, cooking kettles, steam lines, air ducts etc.

Liquid-In-Metal Thermometer: the distinct disadvantages of liquid-glass-thermometers are overcome in liquid-in-metal thermometers. A liquid-in-metal is shown in fig 1.5. In which mercury has been used liquid and the metal is steel. This mercury-in-steel thermometer works on exactly the same principle as the liquid-in-glass thermometer. The glass bulb is replaced by a steel bulb



$$S \propto \frac{l^2 \Delta T}{t}$$

where; S – distance moved

l - Length of the metal strip

ΔT – total change in temperature

t - Thickness of metal

The movement of bimetallic strip is utilized to deflect a pointer over a graduated scale. The longer the length of the strip, the larger the deflection of the tip of the strip, since the distance is proportional to the square of the length. A longer can be contained in a relatively small space if a spiral form, the spiral coil is tightened with increase in temperature. As it coils, the counter post rotates clockwise, and thus a pointer attached to the post also moves on a calibrated temperature scale. (Fig 1.2). This type of temperature indicator is often used homes and offices to indicate ambient air temperature.

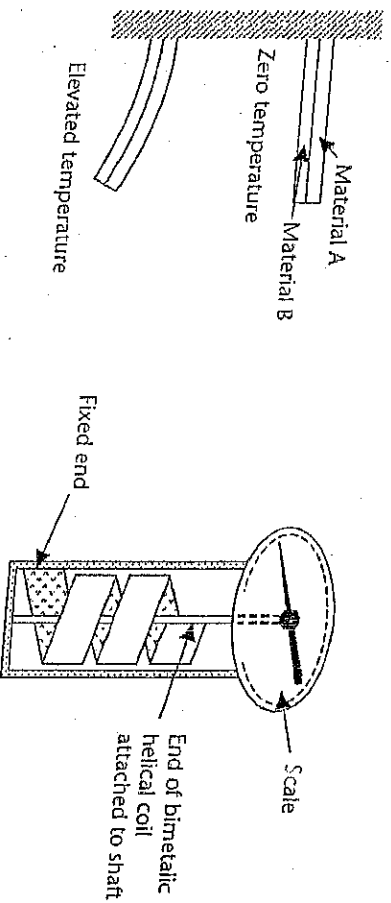


Fig 26: thermometer with spiral helical element.

The bimetal can also be used in the form of a helix to indicate temperature. This type of industrial bimetallic thermometer is shown in fig 1.3. It consists of a tightly wound bimetallic strip located inside the stem of the thermometer with one end fastened permanently to the outer casing. A pointer is attached to a center post that extends from the stem to the center of the indicating dial. The pointer moves on the dial to indicate the measured temperature. Bimetallic thermometers are available for temperature range from -75 to 540°C.

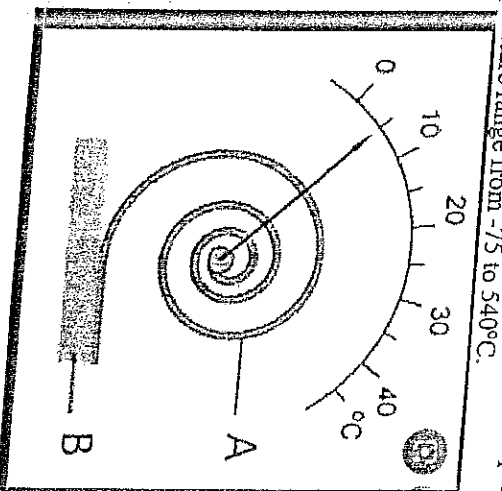


Fig. 27: thermometer with spiral bimetallic element.

Fahrenheit and centigrade (Celsius) temperature scales: the Fahrenheit scale, abbreviated °F was introduced in 1709 by a German philosopher Fahrenheit. On this scale, the melting point of ice is designated as 32°F and the boiling point at 212°F. The centigrade scale was introduced in 1742 by a Swedish astronomer Celsius. On this scale, the ice point of ice is 0°C and the boiling point at 100°C between these two intervals, the Fahrenheit scale is divided into 180 equal divisions and the centigrade scale into 100 equal divisions. Since both scales are linear, temperature can be easily converted from one to other, using the following equation:

$$\frac{^{\circ}\text{C}}{100} = \frac{^{\circ}\text{F} - 32}{180}$$

Kelvin and Rankine temperature scales: the Kelvin scale abbreviated °K was introduced about 1848 by Lord Kelvin. On the Kelvin scale the ice point is 273.15°K and the steam point is 373.15°K the Kelvin scale like the centigrade scale is also divided into 100 equal divisions. The centigrade (°C) can be converted into Kelvin (°K) using the equation:

$$^{\circ}\text{K} = ^{\circ}\text{C} + 273.15$$

The Rankine scale: On the Rankine scale, which is abbreviated as °R, the ice point is 491.7°R and the steam point is 671.7°R. The Rankine scale, like the Fahrenheit scale is divided into 180 equal divisions. Temperatures in Fahrenheit (°F) can be converted into Rankine (°R) by using the equation:

$$^{\circ}\text{R} = ^{\circ}\text{F} + 459.69$$

Both Rankine and Kelvin temperature scales are called the absolute scales because they use absolute zero as one of their reference points.

The Reaumur scale: the Reaumur scale abbreviated as °R° was introduced at about 1731. It assigns 0°R° as the ice point and 80°R° as the steam point. This often used in the alcohol industries Methods of temperature measurement

The temperature measuring instruments are classified according to the nature of change produced in the testing body by the change of temperature. They may be classified as follows:

- Expansion thermometers
- Filled system thermometers
- Electrical temperature instruments
- Pyrometers.

1.3.1 Expansion thermometers: these thermometers are classified according to the substance which expands. They may be described under three headings as follows:

expansion of solids: a) bimetallic thermometers

expansion of liquids: a) liquid-in-glass thermometers

b) liquid-in-metal thermometers

iii) Expansion of gases: a) gas thermometers

Bimetallic Thermometers: the expansion of solids is employed mainly in bimetallic elements by utilizing the difference in thermal expansion of bonded strip of two metals. It consists of two strip of metals invar and brass welded together; each strip made from a metal having a different coefficient of thermal expansion. Whenever the welded strip is heated, the two metals will change different length as the temperature rises. This forces the bimetallic strip to bend towards the side of low co-efficient of thermal expansion. If one end of the bimetallic strip is fixed so that it cannot move, the distance the other end moves is directly proportional to the square of the length of the strip as well as the total change in temperature, and inversely proportional to the thickness of the metal.

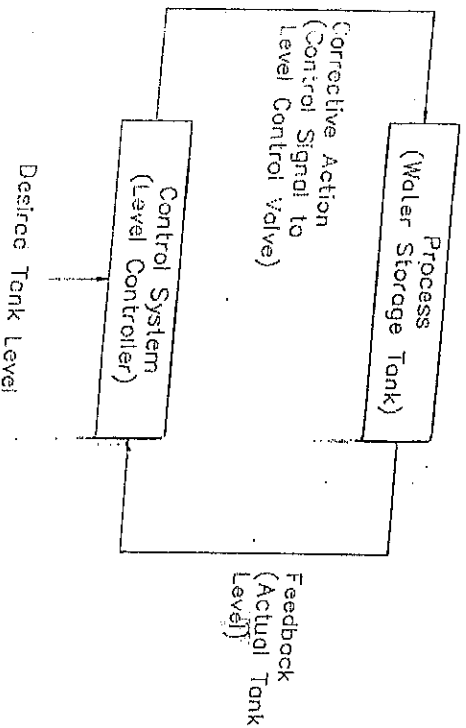


Fig. 58: a feedback control system
A closed-loop control is used to achieve the desired system response using a controller with the output measurement as a feedback signal. The use of feedback enables us to improve system performance at the cost of introducing the measurement noise and stability problem.

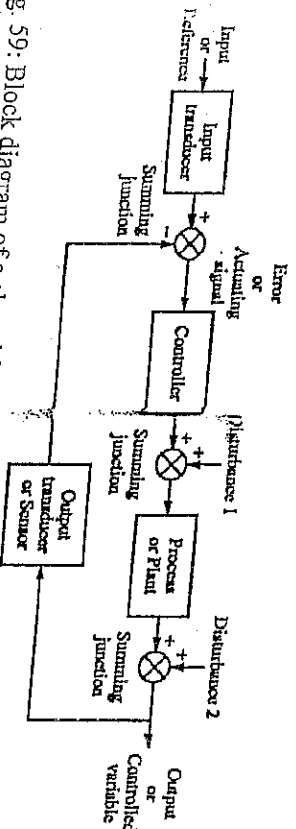


Fig. 59: Block diagram of a closed-loop control system
Control systems that have the property of measurement and correction of disturbances are called closed-loop (feedback) control systems. The generic architecture of a closed-loop system is shown in Figure 59.

The input transducer converts the form of the input to the form used by the controller. For example, if the input to the form used by the controller. For example, if the controller uses electrical signals to operate the valves of a resistor, and the output temperature can be converted to a voltage by a potentiometer, a device whose electrical resistance changes with temperature. The closed-loop system compensates for disturbances by measuring the output response, feeding back through a feedback path, and comparing that response to the input at the summing junction. If there is any difference between the two responses, the system drives the plant, via the actuating signal, to make a correction. If there is no difference, the system drives the plant, since the plant's response is already the desired response.

The following are the advantages of a closed-loop control system:

- In these systems, accuracy is very high due to the correction of any arising error.
 - Since these systems sense environmental changes as well as internal disturbances, the errors are modified.
 - There is a reduced effect of nonlinearity in these systems.
 - Systems have high bandwidth i.e. high operating frequency zone.
 - There are facilities for automation in these systems.
- Disadvantages of Closed-Loop Control System**
- The following are the disadvantages of a closed-loop control system:
- Systems are complicated in design and hence, costlier.
 - Systems may be unstable

Block diagram A pictorial representation of the functions performed by each component and of the flow of signals.

- (1) Safety: Prevent injury to plant personnel, protect the environment by preventing emission and minimizing waste and prevent damage to the process equipment.
- (2) Maintain product quality (composition, purity, color, etc.) on a continuous basis and with minimum cost.
- (3) Maintain the plant production rate at minimum cost. So, we can say that the reasons for automation of process plants are to provide safety and at the same time maintain desired product quality, high plant throughput, and reduce demand for human labor.

Controls are classified with respect to

- technique involved to perform control (*i.e.* human/machines): **manual/automatic control**
- Time dependence of output variable (*i.e.* constant/changing): **regulator/servo**, (also known as regulating/tracking control)
- fundamental structure of the control (*i.e.* the information used for computing the control):

Open-loop/feedback control, (also known as open-loop/closed-loop control):

Types of control systems

Control systems are classified by the control action, which is the quantity responsible for activating the control system to produce the output. The two general classifications are open-loop and closed-loop control systems.

An **open-loop control system** is one in which the control action is independent of the output. An example of an open-loop control system is a chemical addition pump with a variable speed control (Figure 56). The feed rate of chemicals that maintain proper chemistry of a system is determined by an operator, who is not part of the control system. If the chemistry of the system changes, the pump cannot respond by adjusting its feed rate (speed) without operator action.

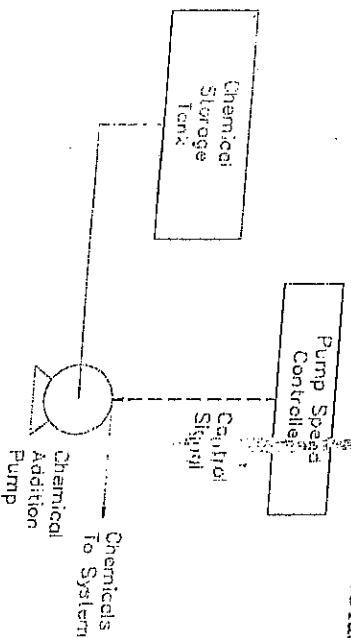
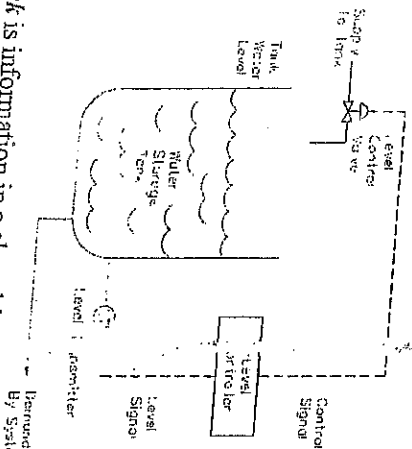


Fig. 56: an open loop control system

A **closed-loop control system** is one in which control action is dependent on the output. Figure 2 shows an example of a closed-loop control system. The control system maintains water level in a storage tank. The system performs this task by continuously sensing the level in the tank and adjusting a supply valve to add more or less water to the tank. The desired level is preset by an operator, who is not part of the system.



Feedback is information in a closed-loop. Feedback in a Closed-Loop Control System control system about the condition of a process variable. This variable is compared with a desired condition to produce the proper control action on the process. Information is continually "fed back" to the control circuit in response to control action.

- Identify and define measurement instrument for calibration.
- Calibrate to prescribed intervals and to national and international standards. (ILAB, NAMAS, NIST).
- Where no standards exist, document the basis used for calibration. (e.g. Test Fixtures, Autocollimators, etc.)
- Define the calibration process and specify the acceptance criteria.
- Label measurement instruments with a suitable indicator to show calibration status.
- Publish calibration status reports on defined intervals.
- When measurement instruments are not conforming, take appropriate action on same and product affected.
- Maintain calibration records.
- Protect measurement instruments from damage, misuse and deterioration. (Use suitable packaging when not in use)
- Safeguard measurement instruments, including software and hardware, from adjustments which would invalidate the calibration setting.

necessary steps for the calibration of instruments

- Examine the construction of the instrument and identify and list all possible inputs
- Decide, as best as one can, which of the inputs will be significant in the application for which the instrument is to be calibrated
- Procure apparatus that will allow all significant inputs to vary over a range considered necessary
- By holding some inputs constant, varying others, and recording the output, develop the desired input output relation.

Control system and its application
System: A system is a combination or an arrangement of different physical components which act together as an entire unit to achieve a certain objective.

A system is a collection, set, or arrangement of elements (subsystems). A kite is an example of a physical system because it is made of paper and sticks. A classroom and a lamp are also examples of the physical system. Systems include physical, biological, organizational, and other entities, and combinations thereof, which can be represented through a common mathematical symbolism. The study of feedback control systems is essentially a study of an important aspect of systems engineering and its application.

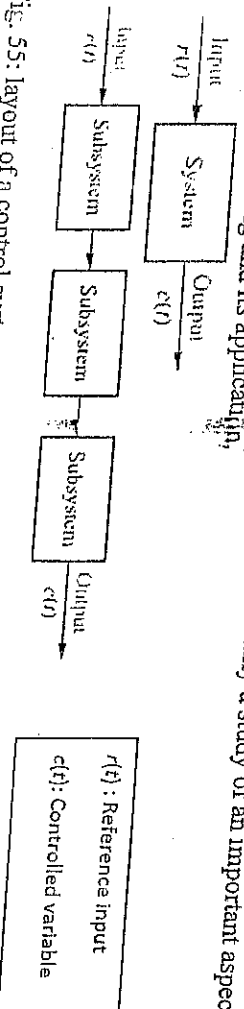


Fig. 55: layout of a control system

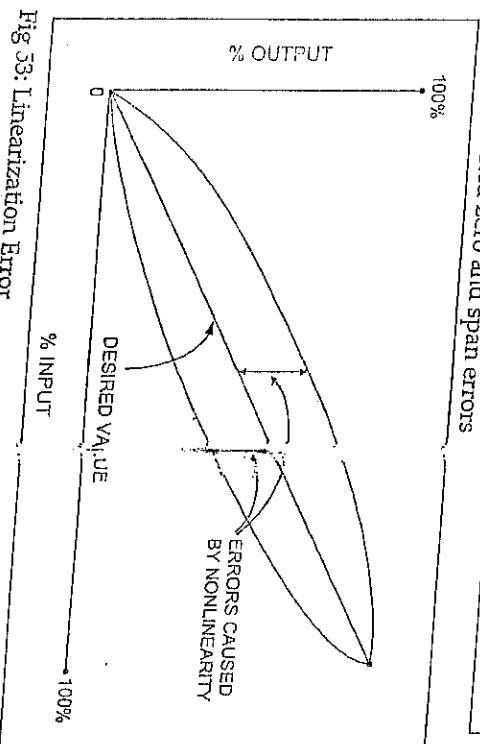
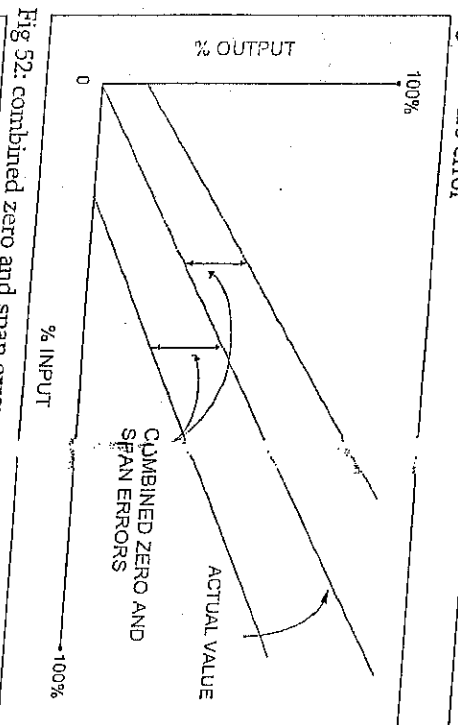
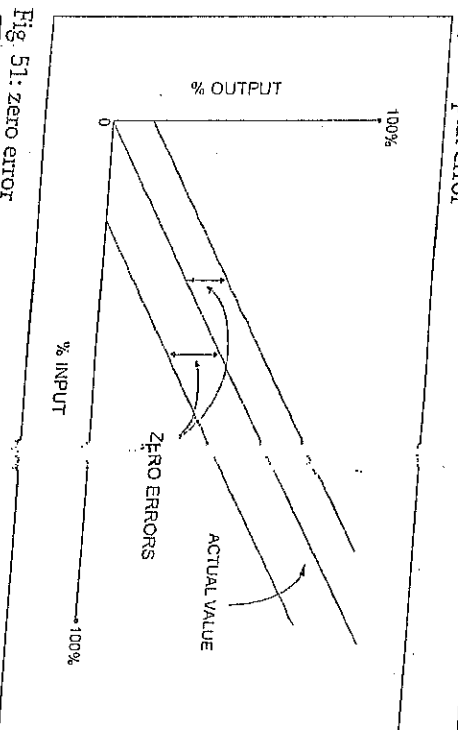
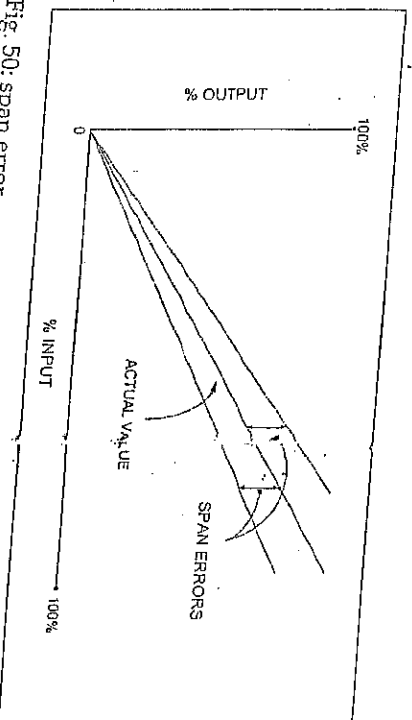
- Input:
- The input is the stimulus, excitation or command applied to a control system.
 - The input is an applied signal or an excitation signal applied to a control system.
- Output:
- The output is the actual response obtained from a control system when an input is applied.
 - It may or may not be equal to the specified response implied by the input.

Plant or Process: The portion of a system which is to be controlled or regulated is called the plant or the process.

Control: The meaning of control is to regulate, direct or command a system so that the desired object is obtained. Control is a set of technologies that achieve desired patterns of variations of operational parameters and sequences for machines and systems by providing the necessary input signals. The control means to regulate, direct, command, or govern.

A control system is a system of integrated elements whose function is to maintain a process variable at a desired value or within a desired range of values. The control system monitors a process variable or variables, then causes some action to occur to maintain the desired system parameter.

Importance of control systems



Calibration requirements include the need to...

- Establish and maintain documented procedures.
- Determine measurements to be made and accuracy required.
- Select an appropriate measurement instrument capable of measurement accuracy and precision.

- ii. **MATLAB:** is a programming platform designed specifically for engineers and scientists to analyze and design systems and products that transform our world. The heart of MATLAB is the MATLAB language, a matrix-based language allowing the most natural expression of computation and mathematics.
 - iii. **Measurement studio:** NI Measurement Studio is a set of test and measurement components built by National Instruments, that integrates into the Microsoft Visual Studio environment. It includes extensive support for accessing instrumentation hardware. It has drivers and abstraction layers for many different types of instruments and buses are included or are available for inclusion.
 - iv. **Scilab:** Scilab is an engineering tool. You can do a lot of things with it. It is capable of numerical computations, data analysis and plotting.
 - v. **orCADspice:** The name is an acronym for Personal Simulation Program with Integrated Circuit Emphasis. It typically takes a netlist generated from orCAD Capture, but can also be operated from MATLAB/Simulink. PSpice lets you simulate and analyze your analog and mixed-signal circuits within orCAD.
 - vi. **GNU Octave:** GNU Octave is a high-level programming language primarily intended for scientific computing and numerical computation. Octave helps in solving linear and nonlinear problems numerically, and for performing other numerical experiments using a language that is mostly compatible with MATLAB.
- Principle of Industrial Calibration

Calibration is the activity of checking, by comparison with a standard, the accuracy of a measuring instrument of any type. It may also include adjustment of the instrument to bring it into alignment with the standard. Even the most precise measurement instrument is of no use if you cannot be sure that it is reading accurately.

Definitions:

Calibration Range: the region between the within which a quantity is measured, received, or transmitted which is expressed by stating the lower and upper range values.

Zero Value: The lower end of the calibration range.

Span: the difference between the upper and lower range.

Instrument Range: the capability of the instrument (may be different than the calibration range).

CHARACTERISTICS OF CALIBRATION

Measurement Instrument: Any instrument which monitors or controls a critical parameter of a manufacturing process, controlled environment or used to measure a product or component specification.

Calibration Tolerance: Every calibration should be performed to a specified tolerance. The terms tolerance and accuracy are often used incorrectly. In ISA's *The Automation, Systems, and Instrumentation Dictionary*, the definitions for each are as follows:

Accuracy: The ratio of the error to the full-scale output or the ratio of the error to the output, expressed in percent span or percent reading, respectively.

Tolerance: Permissible deviation from a specified value; may be expressed in measurement units, percent of span, or percent of reading.

Standard: A defined reference tool with traceability to a national standard, representing the country which operates that laboratory.

Traceability: Documented reference of calibration results to a recognized standard.

Loop Calibration: The calibration of measurement instruments "as installed" in a total system and represents calibration of the instruments "as used".

Resolution: The power of discrimination of an instrument.

Range: The breadth or span of an instrument's capability of measurement.

Instrument error can occur due to a variety of factors: drift, environment, electrical supply, addition of components to the output loop, process changes, etc. Since a calibration is performed by comparing or applying a known signal to the instrument under test, errors are detected by performing a calibration. An error is the algebraic difference between the indication and the actual value of the measured variable.

Typical errors that occur include:

Scanning Laser gauge

Photo diode array imaging

Low-frequency laser interferometer
Laser and FPD based

2.2.2. Laser based distance measuring instruments

Principle of Laser

LASER INTERFERENCE

LASER INTERFERED ONCE

- High accuracy

- Long range optical path
- Easy installations
- Wear and tear

...fraction is to the measured.



Fig. 49: laser interferometer

digital form.

 $\vdash$

Labview: LabVIEW is a graphical, powerful and flexible instrumentation and analysis software system for scientists and engineers

6. **Sampling length:** - Length of profile necessary for the evaluation of the irregularities.
7. **Waviness:** - Surface irregularities which are of greater spacing than roughness.

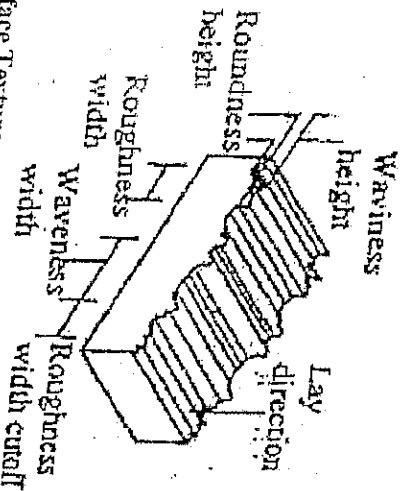


Fig. 48: Surface Texture

8. **Roughness height:** - Rated as the arithmetical average deviation.
9. **Roughness width:** - Distance parallel to the normal surface between successive peaks.
10. **Mean line of profile:** - Line dividing the effective profile such that within the sampling length.
11. **Centre line of profile:** - Line dividing the effectiveness profile such that the areas embraced by profile above and below the line are equal.

Analysis of surface finish

- The analyses of surface finish being carried out by
1. The average roughness method.
 2. Peak to valley height method
 3. Form factor

Methods of measuring surface finish

- The methods used for measuring the surface finish is classified into
1. Inspection by comparison
 2. Direct Instrument Measurements

1. Inspection by comparison methods:

In these methods the surface texture is assessed by observation of the surface. The surface to be tested is compared with known value of roughness specimen and finished by similar machining process.

The various methods which are used for comparison are

1. Touch Inspection.
2. Visual Inspection.
3. Microscopic Inspection.
4. Scratch Inspection.
5. Micro Interferometer.
6. Surface photographs.
7. Reflected Light Intensity.
8. Wallace surface Dynamometer.
2. Direct instrument measurements

2. Direct instrument measurements

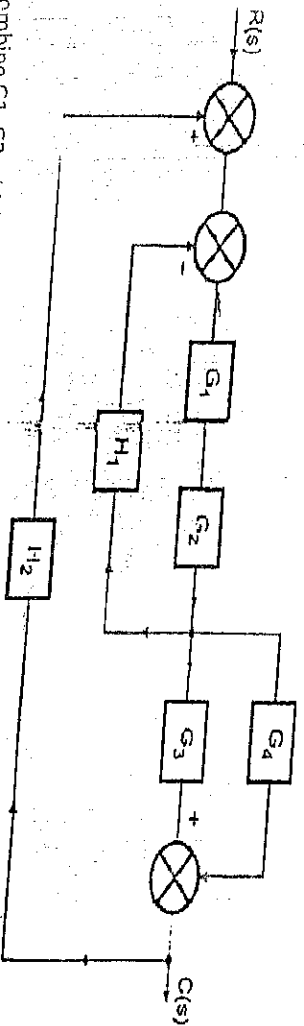
Some of the direct measurement instruments are

1. Stylus probe instruments.
2. Tomlinson surface meter.
3. Profilometer.
4. Taylor-Hobson Talysurf

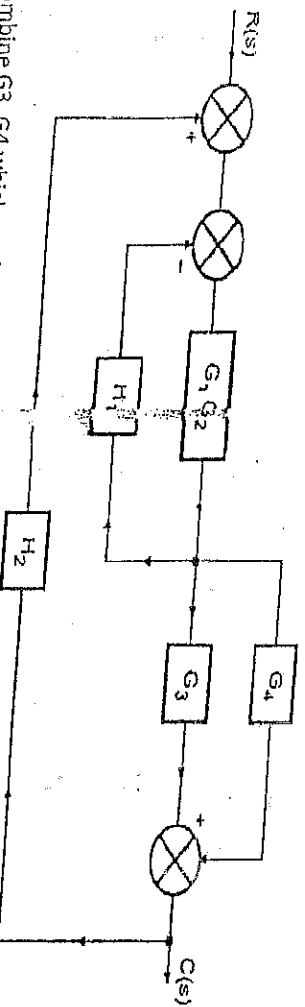
Roundness Measurements

Roundness is defined as a condition of a surface of revolution. Where all points of the surface intersected by any plane perpendicular to a common axis in case of cylinder and cone.

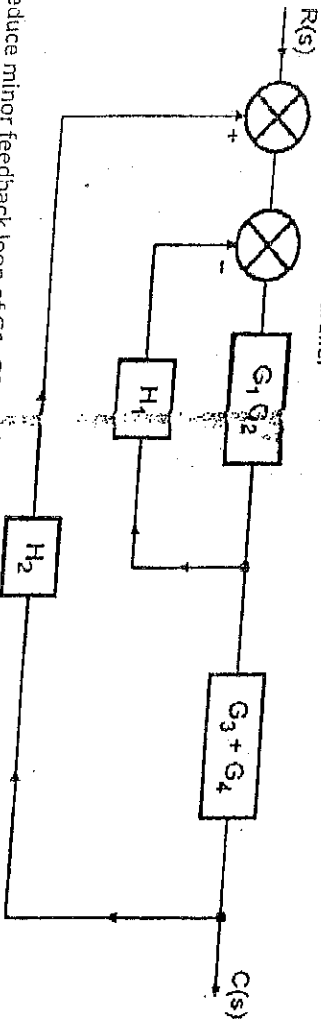
- Devices used for measurement of roundness
- 1) Diameter gauge.
 - 2) Circumferential conferring gauge => a shaft is confined in a ring gauge and rotated against a set indicator probe.
 - 3) Rotating on center
 - 4) V-Block



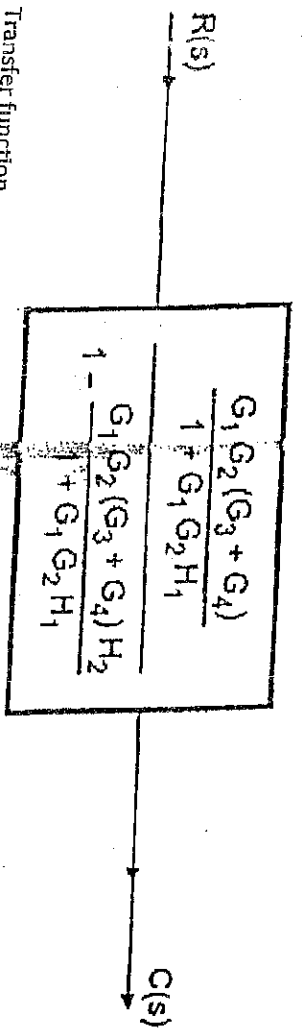
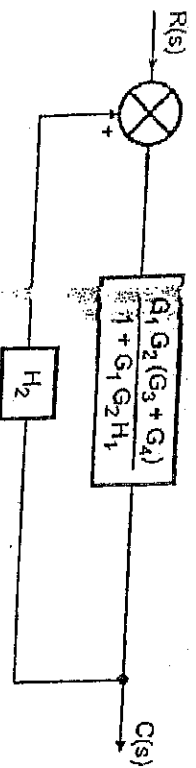
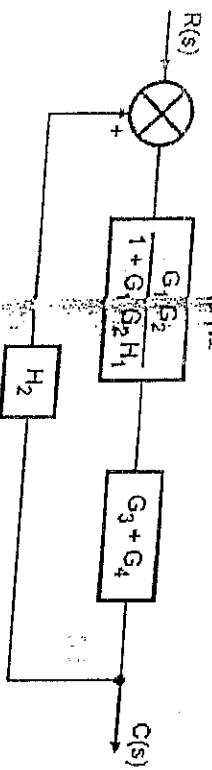
Combine G_1, G_2 which are in series



Combine G_3, G_4 which are in Parallel



Reduce minor feedback loop of G_1, G_2 and H_1



Transfer function

$$\frac{C(s)}{R(s)} = \frac{G_1 G_2 (G_3 + G_4)}{1 + G_1 G_2 H_1 - \frac{G_1 G_2 (G_3 + G_4) H_2}{1 + G_1 G_2 H_1}}$$